

TICVAI

Retail Backend Structure Module

Functional Reference & Dashboard Screen Guide

Configuration • Catalog • Sales • Inventory • Commerce • Analytics

REFERENCE DOCUMENT

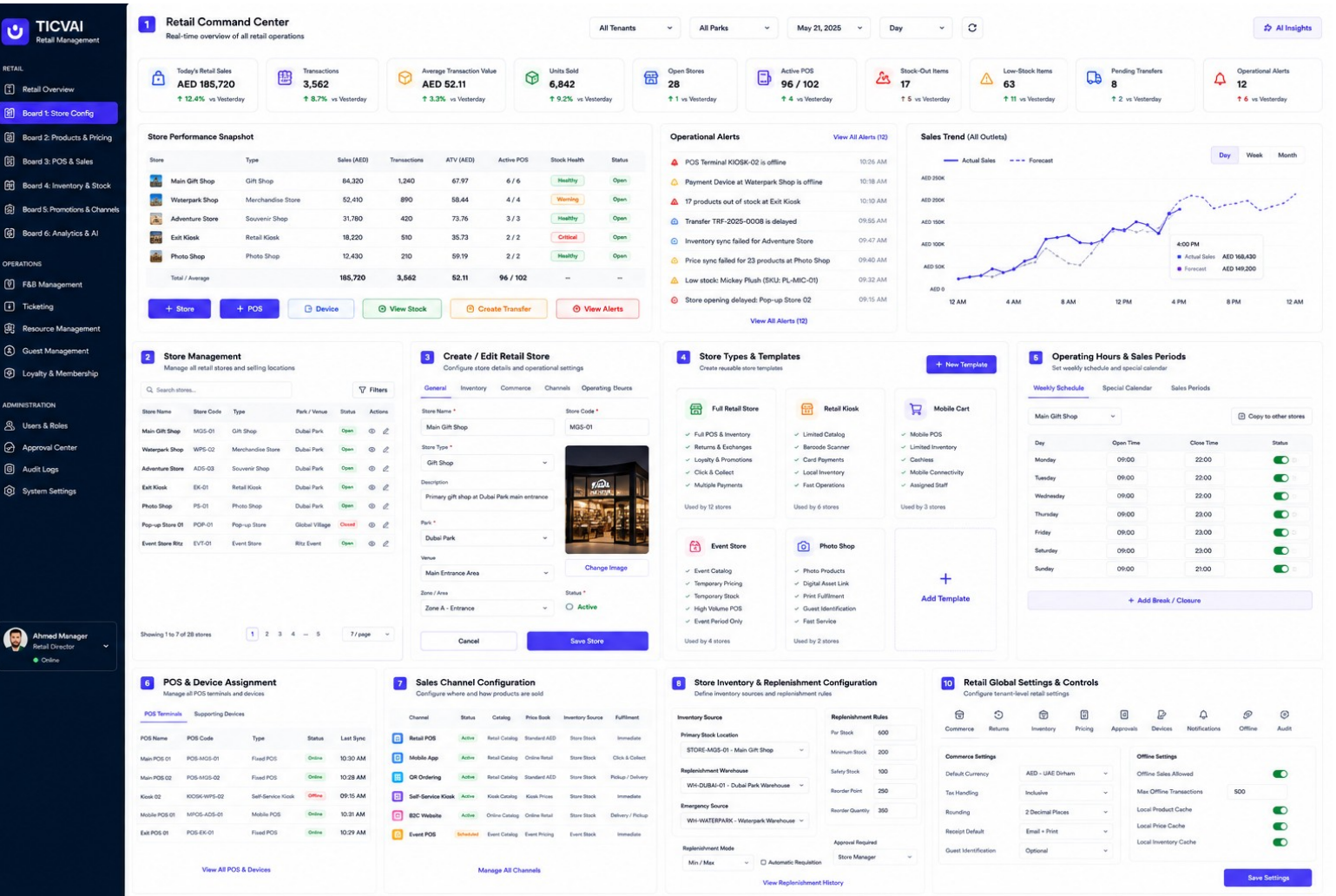
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TICVAI RETAIL — BOARD 1

• Retail Store & Operational Configuration — Scope of Work



1. Purpose

The Retail Store & Operational Configuration Board shall provide the centralized administrative foundation for configuring and managing all retail operations across TICVAI.

The Board shall allow authorized users to create, configure, activate, monitor and control retail stores, shops, kiosks, carts, pop-up stores and other retail selling locations across multiple tenants, companies, parks and venues.

The architecture shall support:

Tenant → Company → Park → Venue → Retail Store → Zone/Area → POS/Device → Sales Channel

The Board shall establish the operational structure used by all downstream Retail capabilities, including:

- Product Catalog
- Pricing

- Merchandising
- POS
- Inventory
- Stock Transfers
- Promotions
- Returns
- Click & Collect
- Omnichannel Commerce
- Guest Profile
- Loyalty
- Payments
- Finance

Reporting

- AI & Forecasting

Retail outlets shall not operate as isolated systems. They shall consume common TICVAI services including Product, Inventory, Procurement, Payment, Finance, Guest, Loyalty, Promotion, Approval and Audit capabilities.

PAGE 1 — Retail Command Center

Purpose

Provide Retail management with a centralized operational overview of all stores and retail selling locations.

Global Filters

- Tenant
- Company
- Park
- Venue
- Store
- Store Type
- Channel
- Date

Status

KPI Cards

- Today's Retail Sales
- Orders / Transactions
- Average Transaction Value
- Units Sold
- Open Stores
- Active POS
- Stores Offline
- Stock-Out Items
- Low-Stock Items
- Pending Transfers
- Pending Approvals

Operational Alerts

- **Store Performance Snapshot**

Display retail locations with:

Store	Sales	Transactions	ATV	Active POS	Stock Health	Status
Main Gift Shop	AED 84K	1,240	AED 67.7	6/6	Healthy	Open
Waterpark Shop	AED 52K	890	AED 58.4	4/4	Warning	Open
Adventure Store	AED 31K	420	AED 73.8	3/3	Healthy	Open

Store	Sales	Transactions	ATV	Active POS	Stock Health	Status
Exit Kiosk	AED 18K	510	AED 35.3	2/2	Critical	Open

Operational Alerts

Examples:

- POS Terminal Offline
- Payment Device Offline
- Low Stock
- Product Out of Stock
- Transfer Delayed
- Store Opening Delayed
- Price Synchronization Failure
- Inventory Synchronization Failure
- Pending Approval
- Device Sync Failure

Quick Actions

+ Store | + POS | + Device | View Stock | Create Transfer | View Alerts

PAGE 2 — Store Management

Purpose

Provide a master list of all Retail stores and selling locations configured within TICVAI.

Store Types

The system shall support configurable store types such as:

- Gift Shop
- Merchandise Store
- Souvenir Shop
- Retail Kiosk
- Mobile Cart
- Pop-Up Store
- Photo Shop
- Specialty Store
- Event Merchandise Store
- Exit Shop
- Online Store
- Warehouse-Linked Store

Store types shall be configurable rather than hard-coded.

Store List

Each store shall display:

- Store Name
- Store Code
- Store Type
- Tenant
- Company
- Park
- Venue
- Zone
- Cost Center
- Revenue Center
- Currency
- POS Count
- Stock Location
- Sales Channels

Status

Last Synchronization

Actions

View | Edit | Duplicate | Activate | Deactivate | Audit History

Bulk Actions

Authorized users should be able to:

Activate Stores

Deactivate Stores

Assign Products

Assign Price Books

Assign Promotions

Change Operating Hours

Export Store List

PAGE 3 — Create / Edit Retail Store

Purpose

Provide the configuration workflow for creating or maintaining a retail selling location.

- **General Information**

Store Name

Store Code

Store Type

Description

Tenant

Company

Business Unit

Park

Venue

Facility

Zone

Floor / Area

Cost Center

Revenue Center

Currency

Tax Profile

Store Logo

Status

- **Contact Information**

Where applicable:

Store Manager

Phone

Email

Internal Contact

Operational Contact

- **Commerce Configuration**

Configure:

Default Product Catalog

Default Price Book

Promotion Set

Tax Rules

Accepted Payment Methods

Refund Policy

Return Policy

Exchange Policy

Receipt Template

Guest Identification Rules

- **Inventory Configuration**

Assign:

Stock Location

Warehouse

Replenishment Source

Default Transfer Location

Par Stock Profile

Reorder Rules

Stock Count Profile

The actual inventory logic shall remain within TICVAI's shared Inventory Engine.

- **Save Options**

Cancel | Save Draft | Save & Next

PAGE 4 — Store Types & Templates

Purpose

Allow administrators to create reusable Retail operating templates.

Instead of configuring every new store manually, TICVAI shall allow a template to define standard operational behavior.

- **Example Templates**

- **Full Retail Store**

- ✓ POS
 - ✓ Inventory
 - ✓ Returns
 - ✓ Exchanges
 - ✓ Loyalty
 - ✓ Promotions
 - ✓ Click & Collect
 - ✓ Multiple Payment Methods
 - ✓ Stock Transfers

- **Retail Kiosk**

- ✓ POS
 - ✓ Limited Catalog
 - ✓ Barcode Scanner
 - ✓ Card Payment
 - ✓ Local Stock
 - ✗ Complex Returns

- **Mobile Cart**

- ✓ Mobile POS
 - ✓ Limited Inventory
 - ✓ Cashless
 - ✓ Mobile Connectivity
 - ✓ Assigned Staff

- **Event Merchandise Store**

- ✓ Event-Specific Catalog
 - ✓ Temporary Pricing
 - ✓ Temporary Stock
 - ✓ High-Volume POS
 - ✓ Event Operating Period

- **Photo Shop**

- ✓ Photo Products
 - ✓ Digital Asset Link

- ✓ Print Fulfilment
- ✓ Guest / Ticket Identification

- **Template Configuration**

Templates may define:

- Product Catalog
- Price Book
- Payment Methods
- Inventory Rules
- Return Rules
- Promotion Rules
- POS Layout
- Receipt Rules
- Operating Hours
- Required Devices
- Sales Channels

- **Template Usage**

The system shall show:

- **Template → Number of Stores Using Template**

Changes to templates shall support controlled propagation rather than automatically changing active stores without authorization.

PAGE 5 — Operating Hours & Sales Periods

Purpose

Configure when each retail location is operational and when specific selling rules apply.

- **Weekly Operating Schedule**

Configure by:

Monday

Tuesday

Wednesday

Thursday

Friday

Saturday

Sunday

For each day:

Opening Time

Closing Time

Status

Break / Closure Periods

- **Special Operating Calendar**

Support:

Public Holidays

Ramadan

Special Events

Private Events

School Groups

Seasonal Operations

Temporary Closures

Extended Hours

Example:

- **New Year's Eve**

Standard Closing:

- 22:00

Override:

- 02:00

- **Sales Periods**

Allow configurable periods such as:

Morning

Day

Evening

Event Period

Closing Period

These periods may later drive:

Pricing

Promotions

Product Availability

Staffing

Reporting

AI Forecasting

- **Scheduled Activation**

Store configuration changes may have:

- **Effective From**

- and

- **Effective Until**

This allows temporary retail operations without manual reconfiguration.

PAGE 6 — POS & Device Assignment

Purpose

Manage all POS terminals and supporting Retail devices assigned to each selling location.

- **Device Types**

Support:

- Fixed POS
- Mobile POS
- Self-Service Kiosk
- Payment Terminal
- Barcode Scanner
- QR Scanner
- RFID Reader
- RFID Handheld Scanner
- Receipt Printer
- Label Printer
- Cash Drawer
- Customer Display
- Tablet
- Handheld Device

- **Device Registry**

Display:

- Device ID
- Device Name
- Device Type
- Store
- POS
- Assigned Operator
- IP / Network Reference
- Software Version
- Last Sync
- Connectivity

Status

- **Statuses**

- Online
- Offline
- Sync Pending

Maintenance

Disabled

- **POS Detail**

Selecting a POS should display:

Store

POS Type

Assigned Cashier

Cash Drawer

Payment Terminal

Barcode Scanner

Receipt Printer

Customer Display

Software Version

Last Transaction

Last Sync

- **Device Actions**

Assign | Reassign | Disable | Replace | Sync | View Logs

PAGE 7 — Sales Channel Configuration

Purpose

Define where and how Retail products may be sold.

Supported Channels

Retail POS
Mobile POS
Self-Service Kiosk
B2C Website
Guest Mobile App
QR Commerce
Click & Collect
Event Sales
Staff-Assisted Sales
Third-Party Commerce
API Sales

Each channel may be:

Active | Inactive | Scheduled

- **Channel Configuration**

For each channel configure:

Product Catalog
Price Book
Promotion Eligibility
Payment Methods
Inventory Source
Fulfilment Method
Receipt Method
Guest Login Requirement
Maximum Quantity
Sales Restrictions

- **Example**

- **Guest Mobile App**

- Catalog:
Dubai Park Retail
- Price Book:
Standard Retail AED

- Inventory:
Main Gift Shop

Fulfilment:

- **Click & Collect / Delivery to Designated Location**

Payment:

- **Online Payment / Wallet / Loyalty where eligible**
- **Channel-Level Product Control**

Authorized users shall be able to define:

- **Product available on POS but unavailable online**

or:

Event merchandise available only through Event Store + Mobile POS.

PAGE 8 — Store Inventory & Replenishment Configuration

Purpose

Configure how each Retail store connects to the shared TICVAI Inventory Engine.

This screen shall configure the relationship; it shall **not duplicate the Inventory Management module**.

- **Inventory Source**

Example:

- **Main Gift Shop**

Primary Stock Location:

- STORE-MGS-01

Replenishment Warehouse:

- WH-DUBAI-01

Emergency Source:

- Waterpark Warehouse

- **Replenishment Rules**

Configure:

Par Stock

Minimum Stock

Maximum Stock

Safety Stock

Reorder Point

Reorder Quantity

Replenishment Frequency

Lead Time

Automatic Requisition

Approval Requirement

- **Example**

- ORYX Plush Toy

- On Hand:

22

- Minimum:

20

- Par:

60

Recommended Replenishment:

- **38 Units**

- **Replenishment Modes**

Manual

Scheduled

Min/Max

Par Stock

Demand-Based

Forecast-Based

AI Recommended

- **Approval**

Example:

- **Replenishment Request**

- → Store Manager

- → Inventory Manager

- → Warehouse Fulfilment

The common TICVAI Approval Engine shall control approval workflows.

PAGE 9 — Store Rules, Controls & Permissions

Purpose

Configure operational controls at Retail-store level.

- **Sales Controls**

Maximum Quantity per Transaction

Maximum Quantity per Guest

Restricted Products

Age Restrictions

Employee Purchase Rules

Complimentary Item Rules

Manual Price Override

Discount Override

Negative Sale Rules

- **Return Controls**

Configure:

Return Window

Receipt Required

No-Receipt Return

Original Tender Requirement

Exchange Allowed

Store Credit Allowed

Damaged Item Handling

Non-Returnable Products

- **Example**

- **Manual Price Override**

Up to AED 20:

- Supervisor

AED 20–100:

- Store Manager

Above AED 100:

- Retail Manager

- **Stock Controls**

Configure:

Negative Stock Allowed

Transfer Approval

Adjustment Approval

Waste / Damage Approval

Count Variance Tolerance

- **User / Role Controls**

Permissions shall use TICVAI's common RBAC/PBAC engine.

Examples:

- **Cashier**
 - Sell + Basic Return
 - **Supervisor**
 - Void + Discount + Price Override
 - **Store Manager**
 - Store Operations + Stock Adjustments
 - **Retail Director**
 - Cross-Store Management
-
-

PAGE 10 — Retail Global Settings & Controls

Purpose

Provide tenant-level Retail configuration applicable across all stores.

- **Tabs**

Commerce | Returns | Inventory | Pricing | Approvals | Devices | Receipts | Notifications | Offline | Audit

- **Commerce Settings**

Configure:

Default Currency

Tax Handling

Rounding

Guest Identification

Receipt Behavior

Transaction Numbering

Barcode Behavior

- **Return / Refund Settings**

Configure:

Default Return Period

Original Tender Rules

Store Credit

Exchanges

Gift Receipt Returns

Refund Approval Levels

- **Inventory Controls**

Configure:

Negative Inventory

Stock Reservation

Stock Synchronization Frequency

Replenishment Behavior

Count Variance Rules

Transfer Rules

- **Approval Matrix**

Examples:

Action	Threshold	Required Role
Manual Discount	≤ 10%	Supervisor

Action	Threshold	Required Role
Manual Discount	> 10%	Store Manager
Price Override	≤ AED 50	Store Manager
Price Override	> AED 50	Retail Manager
Stock Adjustment	≤ AED 500	Store Manager
Stock Adjustment	> AED 500	Inventory Manager

• Offline Configuration

Because TICVAI may operate in attractions where connectivity can be interrupted, Retail shall support configurable offline policies.

Configure:

- Offline Sales Allowed
- Offline Payment Rules
- Maximum Offline Transaction
- Local Product Cache
- Local Price Cache
- Local Inventory Cache
- Synchronization Rules
- Conflict Handling
- Offline Transaction Queue

Notifications

Configure alerts for:

- Store Offline
- POS Offline
- Payment Device Offline
- Inventory Sync Failure
- Price Sync Failure
- Low Stock
- Stock-Out
- Pending Transfer
- Pending Approval
- High Return Rate
- Unusual Discount Activity

• CROSS-MODULE ARCHITECTURE

Board 1 shall configure Retail operations while consuming TICVAI shared services.

The architecture should follow:

- **Retail Store**
- ↓
- **Sales Channels & POS**
- ↓
- **Product / Catalog / Pricing**
- ↓
- **Shared Inventory Engine**
- ↓
- **Shared Procurement Engine**
- ↓
- **Shared Promotion Engine**
- ↓
- **Payment / Wallet / Loyalty**
- ↓
- **Guest Profile**
- ↓
- **Finance**
- ↓
- **Analytics & AI**

This distinction is important.

For example, a store may configure:

Replenish from Warehouse A when stock reaches minimum.

But the actual warehouse stock, transfer transaction, goods movement and valuation should be managed by the **Shared Inventory Engine**.

Similarly, Retail may consume a supplier cost, but supplier quotation comparison, purchase requisition, PO, receiving and supplier invoice workflows should belong to the later **Shared Procurement module**.

• **Recommended Retail Architecture**

One architectural principle should remain consistent with what we did for F&B:

Do not build Retail as another isolated POS + stock system.

The same product sold from a **Gift Shop POS, Mobile POS, Guest App, B2C website, event merchandise stand**, or **self-service kiosk** should use a common commerce architecture:

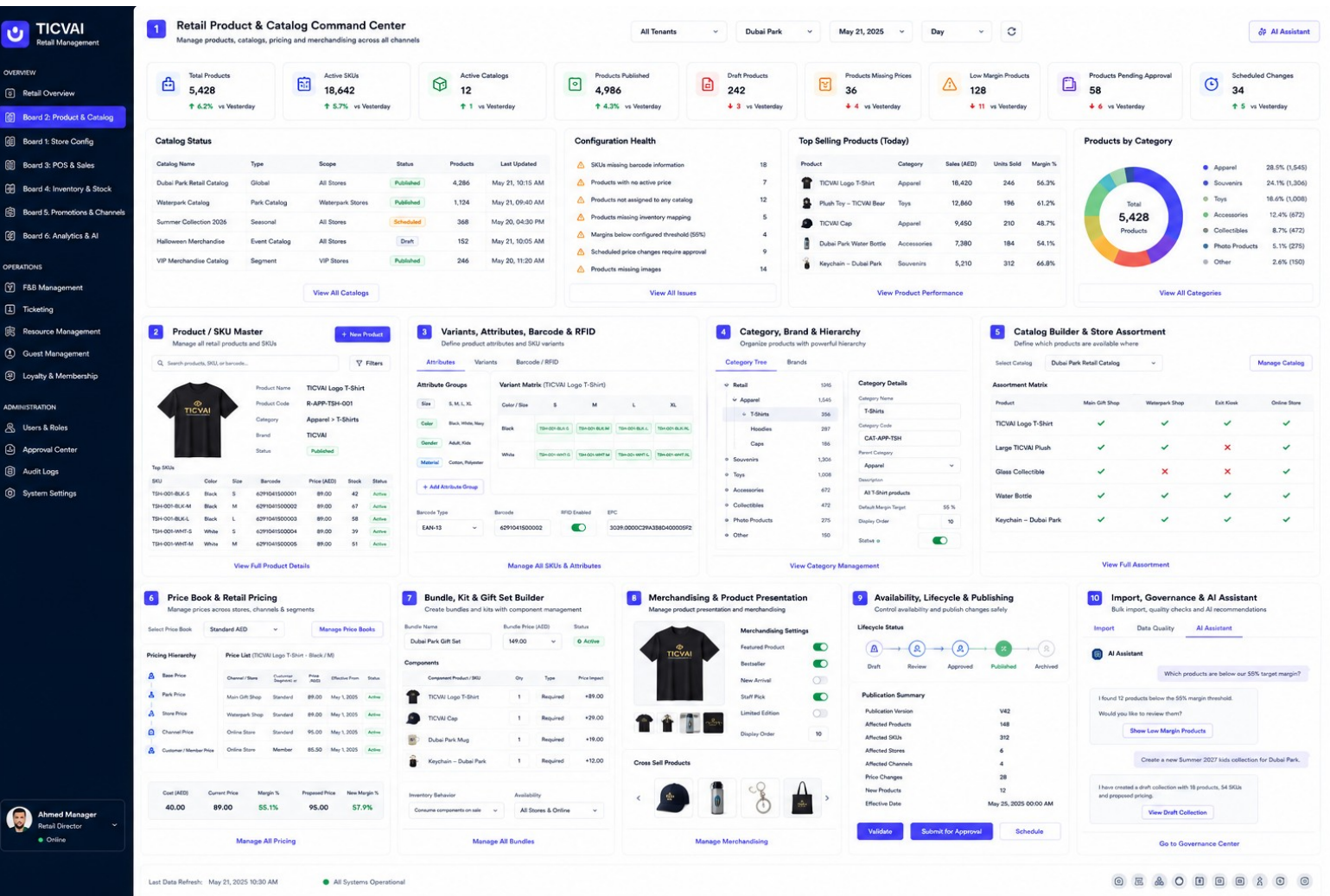
**Product → SKU/Variant → Catalog → Price → Promotion → Channel → Store → Inventory
Location → Sale → Fulfilment → Return → Finance**

Then Retail adds its specialized capabilities such as merchandising, barcode/RFID, variants, returns/exchanges, store stock and omnichannel fulfilment.

This will become particularly important when we reach **Shared Inventory & Procurement**, because F&B and Retail should both consume that same central inventory/procurement architecture rather than maintaining two separate stock systems.

TICVAI RETAIL — BOARD 2

Product, Catalog, Pricing & Merchandising Management — Scope of Work



1. Purpose

The Product, Catalog, Pricing & Merchandising Management Board shall provide the centralized Retail product-management layer for creating, organizing, pricing, merchandising, publishing and controlling all retail products sold through TICVAI.

The Board shall support physical merchandise, souvenirs, apparel, accessories, toys, collectibles, photos, personalized products, event merchandise, bundles and other configurable retail product types.

The core structure shall follow:

Product → SKU / Variant → Attributes → Barcode / RFID → Catalog → Price → Promotion Eligibility → Channel → Store → Availability → Publication

The objective is to maintain **one centralized Retail Product Master**, while allowing different stores, parks, venues and digital channels to determine where, when and at what price each product is sold.

The Board shall integrate with TICVAI's shared:

Inventory Engine

Procurement Engine

Promotion Engine

Payment Engine

Loyalty & Membership

Guest Profile

Finance

Approval Engine

Audit Engine

Analytics & AI

PAGE 1 — Retail Product & Catalog Command Center

Purpose

Provide Retail management with a centralized overview of product configuration, catalogs, pricing, merchandising and publication health.

Global Filters

Tenant
Company
Park
Venue
Store
Catalog
Category
Brand
Product Status
Channel

KPI Cards

Total Products
Active SKUs
Active Catalogs
Products Published
Draft Products
Products Missing Prices
Products Missing Images
Products Missing Barcodes
Products Without Inventory Mapping
Low-Margin Products
Products Pending Approval
Scheduled Changes

- **Catalog Status**

Example:

- **Dubai Park Retail Catalog** — Published
- **Waterpark Catalog** — Published
- **Summer Collection 2026** — Scheduled
- **Halloween Merchandise** — Draft
- **VIP Merchandise Catalog** — Published

- **Configuration Health**

TICVAI shall identify configuration gaps such as:

18 SKUs missing barcode information.

7 products have no active price.

12 products are not assigned to a catalog.

5 products are missing inventory mapping.

4 products have margins below the configured threshold.

9 scheduled price changes require approval.

- **Product Performance Snapshot**

Display:

Product

Category

Selling Price

Cost

Margin

Available Stock

Sales

Status

Quick Actions

+ Product | + Catalog | + Category | + Price Book | Import Products | Publish Changes

PAGE 2 — Product / SKU Master

Purpose

Provide the central Retail Product Master used by all retail channels and locations.

A distinction shall be maintained between:

- **Product** — Commercial parent item
- and

SKU — Specific sellable/inventory variant.

Example:

- **Product**
 - TICVAI Logo T-Shirt
- **SKUs**
 - Black / Small
 - Black / Medium
 - Black / Large
 - White / Small
 - White / Medium
 - White / Large

Each SKU may have its own:

SKU Code
Barcode
RFID/EPC
Cost
Price
Inventory
Weight
Dimensions

Status

- **Product Information**

Capture:

Product Name
Product Code
SKU
Short Name

Description
Product Type
Category
Subcategory
Brand
Collection
Season
Supplier Reference
Manufacturer Reference
Country of Origin
Tax Classification
Product Images
Product Video, where applicable

Status

• Product Types

Configurable examples:

Standard Product
Apparel
Footwear
Souvenir
Toy
Collectible
Personalized Product
Photo Product
Digital Product
Bundle
Gift Item
Event Merchandise

• Commercial Information

Standard Selling Price
Standard Cost
Currency
Tax
Margin
Revenue Account
Cost Account
Revenue Center

Cost Center

- **Physical Information**

Weight

Length

Width

Height

UOM

Pack Size

Case Size

- **Lifecycle Status**

Draft → Review → Approved → Published → Suspended → Discontinued → Archived

PAGE 3 — Variants, Attributes, Barcode & RFID Management

Purpose

Manage complex product variations and product-identification methods without requiring separate manually maintained products for every configuration.

- **Attribute Groups**

Examples:

- **Apparel**

Size

Color

Gender

Fit

Material

- **Souvenir**

Design

Color

Edition

- **Collectible**

Series

Edition

Number

Rarity

- **Variant Matrix**

Example:

- **TICVAI T-Shirt**

Color	S	M	L	XL
Black	SKU-001	SKU-002	SKU-003	SKU-004
White	SKU-005	SKU-006	SKU-007	SKU-008

The system should allow automatic generation of SKU combinations from selected attributes.

- **SKU Generation**

Authorized users may define naming patterns.

Example:

- **DXB-APP-TSH-BLK-M**

Where:

- DXB = Park
- APP = Apparel

- TSH = T-Shirt
- BLK = Black
- M = Medium
- **Barcode Management**

Support:

EAN

UPC

Code 128

QR

Internal Barcode

Supplier Barcode

A SKU may support multiple barcodes where required.

- **RFID**

Where RFID is deployed, capture:

RFID Enabled

EPC

Tag Type

Tag Status

Last Read

Last Known Location

The actual RFID stock movement shall be handled by the Shared Inventory Engine.

PAGE 4 — Category, Brand & Merchandise Hierarchy

Purpose

Provide a flexible product hierarchy for merchandising, navigation, reporting and operational control.

- **Example Hierarchy**

- **Retail**

- ↓

- **Apparel**

- ↓

- T-Shirts

- Hoodies

- Caps

- ↓

- Adult / Kids

or:

- **Souvenirs**

- ↓

- Magnets

- Keychains

- Mugs

- Plush Toys

- Collectibles

- **Hierarchy Levels**

TICVAI shall support configurable hierarchy levels rather than hard-coded structures.

Possible dimensions include:

Department

Category

Subcategory

Product Family

Collection

Brand

Season

Theme

Event

- **Category Configuration**

For each category:

Category Name

Code

Parent

Description

Image

Display Order

Tax Profile

Default Margin Target

Default Attributes

Default Return Policy

Default Inventory Behavior

- **Brand Management**

Capture:

Brand Name

Brand Code

Logo

Description

Supplier Relationship

Active Products

- **Seasonal / Themed Collections**

Examples:

- **Summer 2026**
- **Halloween**
- **National Day**
- **Christmas**
- **Formula 1 Weekend**
- **Concert Merchandise**

Products may belong to multiple collections without duplicating the Product Master.

PAGE 5 — Catalog Builder & Store Assortment

Purpose

Define which products are available to specific parks, venues, stores and digital channels.

The Product Master answers:

What is the product?

The Catalog answers:

Where is the product commercially available?

• Catalog Types

Examples:

Global Retail Catalog

Park Catalog

Venue Catalog

Store Catalog

Online Catalog

Event Catalog

VIP Catalog

Employee Catalog

Seasonal Catalog

• Visual Catalog Builder

Authorized users shall be able to:

Search Products

Drag Products

Add Categories

Reorder Categories

Assign Collections

Hide Products

Highlight Products

Copy Catalogs

• Assortment Matrix

Example:

Product	Main Gift Shop	Waterpark	Exit Kiosk	Online
Logo T-Shirt	✓	✓	✓	✓
Large Plush	✓	✓	×	✓
Glass Collectible	✓	×	×	✓

Product	Main Gift Shop	Waterpark	Exit Kiosk	Online
Water Bottle	✓	✓	✓	✓

- **Bulk Assignment**

Support:

- **Assign Collection → 12 Stores**
- **Remove Category → Event Stores**
- **Copy Assortment → New Store**
- **Effective Dates**

Products may have:

Available From

Available Until

This supports temporary merchandise and events.

PAGE 6 — Price Book & Retail Pricing Management

Purpose

Provide centralized configuration of retail selling prices across stores, channels, customer segments and time periods.

- **Price Book Examples**

Standard AED

Standard SAR

Online Retail

VIP Retail

Member Retail

Employee Retail

Event Pricing

Seasonal Pricing

- **Price Record**

Each price may include:

Product / SKU

Price Book

Selling Price

Currency

Tax

Effective From

Effective Until

Store

Channel

Customer Segment

Membership Tier

Status

- **Pricing Hierarchy**

Example:

- Base Price

- ↓

- Park Price

- ↓

- Store Price

- ↓

- Channel Price

- ↓
- Customer / Membership Price

The system shall apply a defined precedence rule where multiple prices are eligible.

- **Margin Visibility**

When changing a price:

- Current Price:
AED 100
- Cost:
AED 45
- Current Margin:
55%
- Proposed Price:
AED 90
- New Margin:
50%

TICVAI should show the impact **before approval**.

- **Bulk Pricing**

Authorized users may:

Increase Category by %
Decrease Collection by %
Copy Price Book
Import Prices
Schedule Price Changes

Example:

Increase all Summer Collection products by 5% from 1 September.

This creates a proposed change set for review.

PAGE 7 — Bundle, Kit & Gift Set Builder

Purpose

Create retail products containing multiple component SKUs while maintaining correct stock consumption.

- **Bundle Types**

Fixed Bundle

Gift Set

Kit

Multi-Pack

Mix & Match

Build Your Own

Promotional Bundle

- **Example**

- **Dubai Park Gift Set — AED 149**

Contains:

- 1 × Logo T-Shirt
- 1 × Cap
- 1 × Mug
- 1 × Keychain

- **Component Rules**

Configure:

Required Component

Quantity

Optional Component

Choice Group

Substitute Allowed

Variant Selection

Price Adjustment

- **Inventory Behavior**

The bundle itself should not create false inventory where component-level stock is required.

Sale:

- **1 × Gift Set**

shall consume:

- **1 × T-Shirt**
- **1 × Cap**
- **1 × Mug**

- **1 × Keychain**

from the Shared Inventory Engine.

- **Availability**

If a mandatory component becomes unavailable, TICVAI shall determine whether the bundle:

- Becomes Unavailable

- Uses Approved Substitute

- Remains Available with Reduced Options

according to configured rules.

PAGE 8 — Merchandising & Product Presentation

Purpose

Control how Retail products are commercially presented across physical and digital selling channels.

- **Digital Product Presentation**

Configure:

Product Name

Short Description

Long Description

Primary Image

Image Gallery

Video

Feature Highlights

Product Specifications

Size Guide

Care Instructions

Search Keywords

Tags

- **Merchandising Controls**

Featured Product

Bestseller

New Arrival

Limited Edition

Exclusive

Recommended

Trending

Staff Pick

Event Merchandise

- **Display Ordering**

Administrators should be able to control:

- **Category Position**

- **Product Position**

- **Featured Placement**

- **Digital Display Order**

- **Cross-Sell**

Example:

Guest selects:

- **TICVAI T-Shirt**

Suggested:

- Cap
- Water Bottle
- Bag
- **Upsell**

Example:

Standard Plush:

- AED 65

Suggested:

- **Limited Edition Plush — AED 95**
- **Related Products**

Relationships may include:

Similar Product

Complementary Product

Alternative

Upgrade

Accessory

Replacement

These relationships shall later feed Board 5 and Board 6's commerce and AI capabilities.

PAGE 9 — Product Availability, Lifecycle & Publishing

Purpose

Control when and where products become available and provide a safe publication workflow for product changes.

- **Availability Rules**

Configure by:

Tenant

Park

Venue

Store

Channel

Date

Time

Inventory

Customer Segment

Membership

Event

Season

- **Product Availability**

Example:

- **Limited Edition F1 Cap**

Available:

- 15–18 November

Locations:

- Event Store ✓
- Main Gift Shop ✓
- Online ✓
- Exit Kiosk ✕

- **Stock-Based Availability**

The system shall support:

- **Available**
- **Low Stock**
- **Out of Stock**
- **Temporarily Unavailable**
- **Discontinued**
- **Publishing Workflow**

Changes shall follow:

- **Draft**
- ↓
- **Validation**
- ↓
- **Approval**
- ↓
- **Scheduled**
- ↓
- **Published**
- **Pre-Publish Validation**

Before publishing, TICVAI should validate:

- ✓ Product Code
- ✓ SKU
- ✓ Barcode
- ✓ Price
- ✓ Tax
- ✓ Inventory Mapping
- ✓ Images
- ✓ Catalog Assignment
- ✓ Channel Assignment
- △ Margin Below Target
- △ Missing Translation
- △ Stock Not Available
- **Publication Summary**

Example:

- **Retail Publication V42**

Affected:

- 148 Products
- 312 SKUs
- 6 Stores
- 4 Channels
- 28 Price Changes
- 12 New Products

Actions

Validate | Submit for Approval | Publish Now | Schedule | Rollback

PAGE 10 — Product Import, Governance & AI Configuration Assistant

Purpose

Provide enterprise-grade tools for bulk product management, governance, quality control and AI-assisted configuration.

- **Bulk Import**

Support structured imports for:

Products

SKUs

Attributes

Barcodes

Categories

Prices

Catalog Assignments

Images / Image References

Supplier References

- **Import Workflow**

- **Upload**

- ↓

- **Map Fields**

- ↓

- **Validate**

- ↓

- **Preview Errors**

- ↓

- **Correct**

- ↓

- **Submit**

- ↓

- **Approval**

- ↓

- **Import**

- **Validation Examples**

SKU 1842 already exists.

Barcode 6291041500213 assigned to another SKU.

17 products have no category.

8 prices are below configured margin threshold.

3 products reference invalid tax profiles.

• **AI PRODUCT CONFIGURATION ASSISTANT**

Authorized users shall be able to use natural language to create or modify product configurations.

Example:

Create a new Summer 2027 kids merchandise collection for Dubai Park. Include T-shirts in four sizes and three colors, caps, water bottles and a gift set. Make the collection available at Main Gift Shop, Exit Shop and online from 1 June to 31 August.

TICVAI should create a **draft configuration**, including:

- Collection
- Product Structures
- Variant Matrix
- Proposed SKUs
- Categories
- Catalog Assignment
- Availability Dates
- Store Assignment
- Channel Assignment

The AI shall not automatically publish.

• **AI Pricing Assistance**

Example:

Which products are below our 55% target margin?

TICVAI identifies products and may propose pricing scenarios.

• **AI Product Data Quality**

TICVAI should identify:

- Duplicate Products
- Duplicate SKUs
- Missing Attributes
- Missing Images
- Missing Prices
- Missing Barcodes
- Incorrect Category Assignment
- Unusual Pricing
- Missing Inventory Mapping

- **PRODUCT VERSIONING & AUDIT**

The system shall maintain version history for significant product changes.

Example:

- **SKU-TSH-001**
- **Version 12**
- Price:
AED 89 → AED 95
- Category:
No Change
- Description:
Updated
- Availability:
Online added
- Changed By:
Retail Manager
- Approved By:
Commercial Director
- Effective:
01 Sep 2026

Users shall be able to:

View Version | Compare Versions | View Approval | Rollback where permitted

- **MULTILINGUAL PRODUCT MANAGEMENT**

Product-facing content shall support multiple languages.

At minimum, the architecture shall support:

English
Arabic

and allow additional tenant-configurable languages.

Fields may include:

Product Name
Description
Category Name
Specifications
Care Instructions
Digital Merchandising Text

The system shall show translation completeness before publication.

Example:

- English:
100%
- Arabic:
92%

Warning:

3 products are missing Arabic descriptions.

• **PERSONALIZED PRODUCTS**

The Product Master shall support configurable personalization where applicable.

Examples:

Name Printing
Jersey Number
Custom Text
Photo Printing
Engraving
Color Selection

• **Example**

• **Personalized Jersey**

- Base:
AED 120

Add Name:

AED 20

Add Number:

AED 15

• **Personalization Data**

Capture:

Personalization Type
Customer Input
Character Limit
Allowed Characters
Additional Price
Production Location
Preparation Time
Fulfilment Method

Personalization information shall be passed to the appropriate fulfilment workflow.

- **LIMITED-EDITION & SERIALIZED PRODUCTS**

TICVAI should support merchandise where individual units may need unique identification.

Example:

- **Limited Edition Collectible**

Edition:

- 500 units

Item:

- **#184 / 500**

The product configuration may include:

Edition Size

Serial Number

Certificate

RFID/EPC

Unique Barcode

Ownership/Guest Link where required

This can later support collectible and premium merchandise use cases.

- **PROMOTION ENGINE BOUNDARY**

Board 2 shall define whether a product is:

Promotion Eligible

Discount Eligible

Coupon Eligible

Loyalty Discount Eligible

Employee Discount Eligible

However, Board 2 shall **not create a separate promotion engine**.

Actual promotion logic such as:

- **Buy 2 Get 1**
- **20% Off**
- **Spend AED 300 Get AED 50**
- **Member Exclusive**

shall use TICVAI's shared Promotion Engine and be addressed more deeply in Retail Board 5.

• INVENTORY BOUNDARY

Board 2 shall configure the SKU's inventory characteristics and mappings.

Examples:

- Inventory Tracked
- Inventory Item
- UOM
- Barcode
- RFID
- Pack Size
- Stock Location Eligibility

But Board 2 shall not maintain:

- Stock Ledger
- Warehouse Quantity
- Goods Receipt
- Transfers
- Stock Counts
- Inventory Valuation

Those functions belong to the Shared Inventory Engine and Retail Board 4's operational inventory interface.

• PROCUREMENT BOUNDARY

Retail Product Master may reference:

- Preferred Supplier
- Supplier Product Code
- Standard Cost
- Lead Time
- Pack Size

However:

- RFQ
- Quotation Comparison
- Purchase Requisition
- Purchase Order
- Goods Receipt
- Supplier Invoice

Supplier Contract Management
shall belong to the shared Procurement module.

- **APPROVAL CONTROLS**

Sensitive changes shall integrate with TICVAI's shared Approval Engine.

Examples:

- **New Product**
 - Retail Manager
 - Commercial Approval
 - **Price Reduction**
 - If margin remains above threshold
 - Retail Manager
 - If margin drops below threshold
 - Finance + Commercial
 - **Product Discontinuation**
 - → Retail Director
 - **Bulk Price Change**
 - → Commercial + Finance
 - **Publication**
 - → Configurable approval workflow
-

- **BOARD 2 — FINAL 10 SCREENS**

#	Screen
1	Retail Product & Catalog Command Center
2	Product / SKU Master
3	Variants, Attributes, Barcode & RFID Management
4	Category, Brand & Merchandise Hierarchy
5	Catalog Builder & Store Assortment
6	Price Book & Retail Pricing Management
7	Bundle, Kit & Gift Set Builder
8	Merchandising & Product Presentation
9	Product Availability, Lifecycle & Publishing

#	Screen
10	Product Import, Governance & AI Configuration Assistant

- **COMPLETE BOARD 2 FLOW**

- **Create Product**

- ↓

- **Assign Category / Brand / Collection**

- ↓

- **Define Attributes**

- ↓

- **Generate SKUs / Variants**

- ↓

- **Assign Barcode / RFID**

- ↓

- **Map Shared Inventory Item**

- ↓

- **Configure Cost + Price**

- ↓

- **Assign Catalog**

- ↓

- **Assign Stores**

- ↓

- **Assign Channels**

- ↓

- **Configure Merchandising**

- ↓

- **Configure Availability**

- ↓

- **Validate**

- ↓

- **Approval**

- ↓

- **Schedule / Publish**

- ↓

POS + Mobile POS + Kiosk + B2C + Guest App synchronized

- ↓

Sales + Inventory + Finance + Analytics consume the same Product/SKU definition

- **ARCHITECTURAL REQUIREMENT**

Retail Board 2 shall use a **single Retail Product & Catalog Engine**.

TICVAI should **not create separate product masters for POS, online, kiosk, mobile or individual stores**.

The required architecture is:

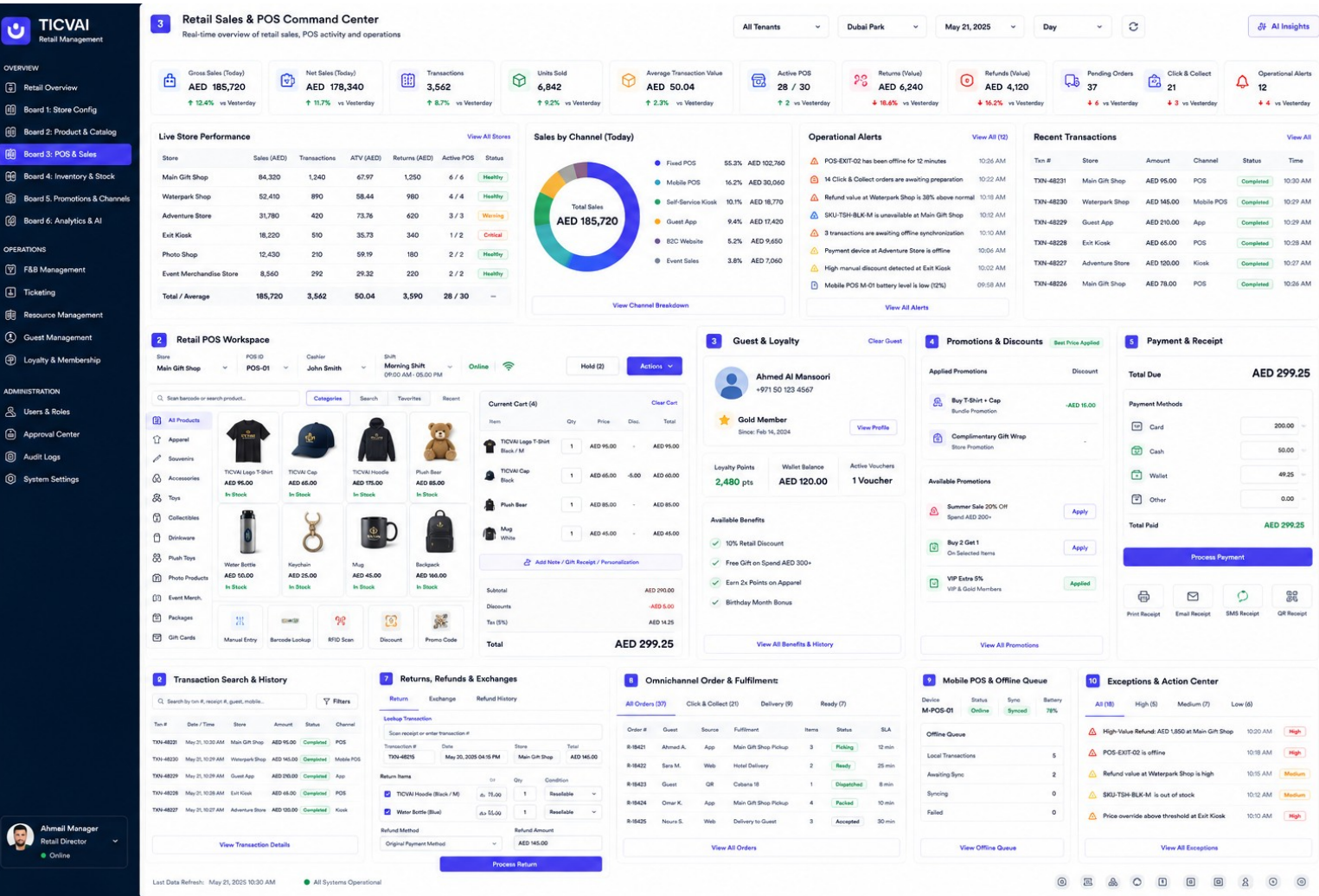
- **Product Master**
- ↓
- **SKU / Variant**
- ↓
- **Attributes + Barcode / RFID**
- ↓
- **Catalog / Assortment**
- ↓
- **Pricing**
- ↓
- **Merchandising**
- ↓
- **Availability**
- ↓
- **Channel Publication**
- ↓

POS / Mobile / Kiosk / B2C / Guest App

This is particularly important for Retail because the same physical SKU may be sold simultaneously through several stores and digital channels while drawing from shared or different inventory locations. A centralized product model will prevent duplicate SKUs, conflicting prices, inconsistent descriptions and fragmented reporting.

TICVAI RETAIL — BOARD 3

Retail POS, Sales, Returns & Fulfilment — Scope of Work



1. Purpose

The Retail POS, Sales, Returns & Fulfilment Board shall provide the complete transaction-execution layer for Retail operations across TICVAI.

The Board shall manage the end-to-end retail transaction lifecycle:

Product Selection → Guest Identification → Price & Promotion → Cart → Payment → Inventory Commitment → Receipt → Fulfilment → Return / Exchange → Financial Posting

The solution shall support high-volume attraction, entertainment, leisure, event and venue retail environments, including:

- Gift Shops
- Merchandise Stores
- Souvenir Shops
- Retail Kiosks
- Mobile Carts

Pop-Up Stores
Event Merchandise Stores
Photo Shops
Specialty Stores
Mobile POS
Self-Service Kiosks
Click & Collect
B2C / Guest App Orders

The objective is to create **one unified Retail Commerce Engine**, rather than separate transaction logic for each sales channel.

All Retail transactions shall integrate in real time, or through controlled offline synchronization, with TICVAI's shared:

Product & Catalog Engine
Pricing Engine
Promotion Engine
Inventory Engine
Payment Engine
Wallet
Loyalty & Membership
Guest Profile
Finance
Approval Engine
Audit Engine
Analytics & AI

PAGE 1 — Retail Sales & POS Command Center

Purpose

Provide Retail management with real-time visibility into transactions, POS activity, fulfilment, returns and operational issues across all stores and channels.

Global Filters

Tenant
Company
Park
Venue
Store
POS
Channel
Date
Shift
Cashier

KPI Cards

Today's Gross Sales
Net Sales
Transactions
Units Sold
Average Transaction Value
Units per Transaction
Active POS
Offline POS
Returns
Refund Value
Exchanges
Pending Fulfilment
Click & Collect Orders

Operational Alerts

• Live Store Status

Store	Sales	Transactions	POS	Returns	Pending Orders	Status
Main Gift Shop	AED 84K	1,240	6/6	18	12	Healthy

Store	Sales	Transactions	POS	Returns	Pending Orders	Status
Waterpark Shop	AED 52K	890	4/4	11	8	Healthy
Adventure Store	AED 31K	420	3/3	6	4	Warning
Exit Kiosk	AED 18K	510	1/2	3	0	POS Offline

• Transaction Channel Breakdown

Display sales by:

- Fixed POS
- Mobile POS
- Self-Service Kiosk
- Guest App
- B2C
- Click & Collect
- Event Sales

• Live Alerts

Examples:

- POS-EXIT-02 has been offline for 12 minutes.
- 14 Click & Collect orders are awaiting preparation.
- Refund value at Waterpark Shop is 38% above normal.
- SKU-TSH-BLK-M is unavailable at Main Gift Shop.
- 3 transactions are awaiting offline synchronization.

Quick Actions

Open POS | Search Transaction | Returns | Fulfilment | Offline Queue | View Alerts

PAGE 2 — Retail POS Selling Workspace

Purpose

Provide a fast, intuitive and configurable selling interface for cashiers and retail employees.

The POS should be optimized for both:

- **Touchscreen operation**
- and

Barcode / RFID-driven high-speed selling.

- **POS Header**

Display:

Store

POS ID

Cashier

Shift

Connectivity

Guest

Current Time

- **Product Entry Methods**

Support:

Barcode Scan

RFID Scan

Product Search

SKU Search

Category Navigation

Product Tiles

Favorites

Recently Sold

Manual Product Entry where authorized

- **Product Tiles**

Tiles may display:

Image

Product Name

Price

Availability

Promotion

Variant Indicator

- **Cart**

Each line shall support:

Product
SKU
Variant
Quantity
Unit Price
Discount
Promotion
Tax
Total

- **Line Actions**

Authorized users may:

- **Change Quantity**
- **Remove Item**
- **Change Variant**
- **Apply Discount**
- **Price Override**
- **Add Note**
- **Gift Receipt**
- **Personalization**
- **Cart Actions**

Hold Transaction
Retrieve Transaction
Clear Cart
Add Guest
Apply Promotion
Apply Loyalty
Apply Voucher
Suspend Sale
Checkout

PAGE 3 — Guest, Membership, Loyalty & Entitlement Interaction

Purpose

Connect Retail transactions with the unified TICVAI Guest Profile while allowing guest checkout when identification is not required.

- **Guest Identification**

Support:

Mobile Number

Email

Membership Card

QR Code

Ticket QR

RFID Wristband

Guest App

Loyalty ID

UAE Pass where configured

Manual Search

- **Guest Summary**

Once identified, POS may display permitted information such as:

- **Guest Name**

- Membership:

Gold

- Loyalty Balance:

2,480 Points

- Wallet:

AED 120

- Available Benefits:

10% Retail Discount

- Active Voucher:

AED 25

- **Guest Checkout Modes**

Support:

Anonymous Guest

Registered Guest

Member

VIP

Group Guest

Employee where permitted

- **Benefit Validation**

The system shall determine applicable:

Member Pricing

Loyalty Discounts

Vouchers

Entitlements

Wallet Credit

VIP Benefits

Employee Benefits

- **Important Architecture**

Retail shall **consume** the common Guest, Loyalty, Membership, Wallet and Entitlement engines.

Board 3 shall not maintain separate guest balances or membership logic.

PAGE 4 — Cart, Promotions, Discounts & Price Control

Purpose

Provide controlled calculation and application of prices, promotions and discounts during the Retail transaction.

- **Pricing Resolution**

The transaction engine shall determine the applicable price based on:

- **SKU**
- ↓
- **Store**
- ↓
- **Channel**
- ↓
- **Price Book**
- ↓
- **Guest / Membership**
- ↓
- **Promotion**
- ↓
- **Manual Adjustment where authorized**
- **Promotion Examples**

Support execution of promotions configured through the common Promotion Engine, including:

Percentage Discount
 Fixed Discount
 Buy X Get Y
 Buy 2 Get 1
 Bundle Price
 Spend Threshold
 Member Exclusive
 Event Promotion
 Coupon
 Voucher
 Promotional Code

- **Promotion Calculation**

Example:

- T-Shirt:
AED 95

- Cap:
AED 65

Promotion:

- **Buy T-Shirt + Cap → AED 145**

Saving:

- **AED 15**
- **Discount Controls**

Manual discounts shall require appropriate permission.

Example:

- 0–10%
- → Supervisor
- 10–20%
- → Store Manager
- Above 20%
- → Retail Manager
- **Price Override**

Display:

- Original Price:
AED 95
- Override:
AED 80
- Difference:
-AED 15
- Reason:
Damaged Packaging
- Approval:
Required
- **Promotion Conflict Resolution**

Where multiple promotions are eligible, TICVAI shall apply configured rules such as:

Best Price
Highest Priority
Stackable
Non-Stackable
Mutually Exclusive
Customer Choice

PAGE 5 — Checkout, Payment & Receipt

Purpose

Complete the Retail transaction through TICVAI's shared Payment Engine.

- **Order Summary**

Display:

Subtotal
Discount
Promotion
Tax
Loyalty Redemption
Voucher
Wallet
Amount Due

- **Payment Methods**

Depending on tenant configuration:

Cash
Credit / Debit Card
Contactless
Apple Pay
Google Pay
TICVAI Wallet
Gift Card
Voucher
Loyalty Points
Account / Credit
Other Integrated Payment Methods

- **Split Tender**

Support combinations such as:

- **AED 100 Card**

- **AED 50 Cash**

- **AED 25 Wallet**

- **Payment Status**

Pending
Authorized

Paid

Partially Paid

Declined

Cancelled

Reversed

- **Receipt Options**

Printed Receipt

Email

SMS

WhatsApp where integrated

Guest App

QR Receipt

No Receipt

- **Gift Receipt**

Allow generation of a receipt without displaying product price where permitted.

- **Transaction Completion**

Once payment succeeds:

- **Payment Confirmed**

- ↓

- **Sale Posted**

- ↓

- **Inventory Reduced / Committed**

- ↓

- **Finance Posting Generated**

- ↓

- **Loyalty Updated**

- ↓

- **Receipt Generated**

- ↓

- **Analytics Updated**

PAGE 6 — Transaction Search, Hold, Recall & Post-Sale Management

Purpose

Provide authorized employees with tools to locate and manage current and historical Retail transactions.

- **Search By**

Transaction Number

Receipt Number

Guest

Membership

Mobile

Email

Date

Store

POS

Cashier

Product

SKU

Barcode

Payment Reference

- **Transaction Detail**

Display:

Transaction Number

Date / Time

Store

POS

Cashier

Guest

Products

Quantity

Prices

Discounts

Promotions

Tax

Payment

Receipt

Fulfilment

Inventory Posting

Loyalty

Return History

- **Transaction Timeline**

Example:

- **14:32:18 — Transaction Started**
- **14:33:04 — Guest Identified**
- **14:34:12 — Promotion Applied**
- **14:35:08 — Payment Authorized**
- **14:35:11 — Sale Completed**
- **14:35:13 — Inventory Posted**
- **Held Transactions**

Cashiers may:

- **Hold → Retrieve → Continue → Complete**

subject to configurable timeout and permission rules.

- **Post-Sale Actions**

Depending on permission:

Reprint Receipt

Send Receipt

Gift Receipt

Start Return

Start Exchange

View Audit

View Payment

View Fulfilment

PAGE 7 — Returns, Refunds & Exchanges

Purpose

Manage controlled product returns, refunds and exchanges across Retail channels.

- **Return Lookup**

Locate original transaction using:

Receipt
Transaction ID
Guest
Barcode
Membership
Order Number

- **Return Validation**

TICVAI shall automatically evaluate:

Return Window
Receipt Requirement
Product Eligibility
Product Condition
Original Purchase
Original Tender
Previous Returns
Non-Returnable Rules
Promotion Impact
Loyalty Impact

- **Return Reasons**

Configurable examples:

Wrong Size
Wrong Product
Changed Mind
Defective
Damaged
Duplicate Purchase
Incorrect Item
Gift Return
Other

- **Product Condition**

Resellable

Damaged
Defective
Opened
Used
Missing Packaging
Write-Off

- **Return Outcome**

Support:

Refund
Exchange
Store Credit
Gift Card
Replacement
No Refund / Reject

- **Exchange Example**

Original:

- T-Shirt Black / M
- AED 95

Exchange:

- T-Shirt Black / L
- AED 95

Difference:

- **AED 0**

or:

- Original:
AED 95
- New Product:
AED 120

Guest Pays:

- **AED 25**
- **Refund Methods**

Subject to payment policy:

Original Tender
Cash
Card
Wallet
Store Credit

Gift Card

Voucher

- **Multi-Tender Refunds**

If the original purchase used multiple tenders, the system shall retain the original tender composition and apply configured refund rules.

- **Inventory Impact**

A returned product shall be routed according to condition:

- **Resellable**

- → Return to Available Stock

- **Damaged**

- → Quarantine / Damaged Stock

- **Defective**

- → Quality / Supplier Workflow

- **Write-Off**

- → Inventory Adjustment Approval
-
-

PAGE 8 — Omnichannel Order & Fulfilment Center

Purpose

Manage Retail orders created outside the physical store and provide one operational fulfilment workspace.

- **Order Sources**

B2C Website

Guest App

QR Commerce

Self-Service Kiosk

Event Commerce

API / Third Party

Assisted Sale

Another Store

- **Fulfilment Types**

Support:

Click & Collect

Store Pickup

Venue Pickup

Hotel / Room Delivery where integrated

Table / Cabana Delivery where applicable

Guest Delivery

Event Pickup

Ship to Address where supported

- **Order Queue**

Order	Guest	Source	Fulfilment	Items	SLA	Status
R-18421	Ahmed	App	Main Gift Shop Pickup	3	12 min	Picking
R-18422	Guest	Web	Hotel Delivery	2	25 min	Ready
R-18423	Sara	QR	Cabana 18	1	8 min	Dispatched

- **Fulfilment Lifecycle**

- **Order Received**

- ↓

- **Inventory Reserved**

- ↓

- **Accepted**

- ↓

- **Picking**
- ↓
- **Packed**
- ↓
- **Ready**
- ↓
- **Guest Notified**
- ↓
- **Collected / Dispatched**
- ↓
- **Completed**
- **Picking**

Employee shall see:

Product

SKU

Variant

Quantity

Stock Location

Bin where applicable

Barcode

Barcode/RFID verification should be supported.

- **Guest Notification**

Examples:

Your order is ready for collection at Main Gift Shop.

or:

Your merchandise is on the way to Cabana 18.

PAGE 9 — Mobile POS, Event Sales & Offline Operations

Purpose

Support Retail selling outside traditional fixed-store environments.

- **Mobile POS Use Cases**

Event Merchandise

Queue-Line Selling

VIP Areas

Temporary Shops

Pop-Up Stores

Mobile Carts

Stadium / Arena Selling

Outdoor Attractions

- **Mobile POS**

The same Retail Commerce Engine shall support:

Product Search

Barcode Scan

RFID Scan

Cart

Guest Identification

Promotions

Payment

Digital Receipt

Inventory Posting

- **Temporary Store Assignment**

Example:

- **Concert Merchandise Stand 04**

- Event:

Summer Live 2026

- Operating:

18:00–00:30

- Catalog:

Concert Merchandise

- Stock:

EVENT-STOCK-04

- POS:

4 Mobile Devices

- **Offline Mode**

Retail shall support controlled offline operation where connectivity is interrupted.

Local cache may include:

- Products
- SKUs
- Barcodes
- Prices
- Tax Rules
- Approved Promotions
- Limited Inventory Snapshot
- Device Configuration
- **Offline Transaction Queue**

Statuses:

- Local
- Awaiting Sync
- Syncing
- Synchronized
- Conflict
- Failed

- **Conflict Management**

Example:

Product sold offline:

- SKU-1842
- Local Inventory:
2
- Central inventory before synchronization:
0

The system shall not silently overwrite the discrepancy.

It shall create:

- **Inventory Synchronization Conflict**

for resolution according to configured policy.

- **Offline Payment Controls**

Tenant configuration shall determine:

- Whether offline payment is allowed
- Maximum transaction value
- Allowed tender types
- Maximum offline duration
- Risk thresholds

PAGE 10 — Sales Exceptions, Controls & Operational Action Center

Purpose

Provide one centralized workspace for managing transaction exceptions, approvals, synchronization issues and unusual Retail activities.

- **Exception Categories**

- **Transaction**

Suspended Sale

Failed Sale

Duplicate Transaction

Negative Sale

Transaction Sync Failure

- **Payment**

Declined Payment

Payment Timeout

Reversal Pending

Tender Mismatch

- **Pricing**

Missing Price

Price Conflict

Unauthorized Discount

Margin Warning

- **Returns**

Return Outside Window

No Receipt

High-Value Refund

Repeated Return

Non-Returnable Product

- **Inventory**

Negative Stock

Stock Conflict

Fulfilment Shortage

Inventory Sync Failure

- **Devices**

POS Offline

Scanner Offline

Payment Terminal Offline

Printer Offline

- **Severity**

Information | Warning | High | Critical

- **Exception Card**

Example:

- **High-Value Refund**

- Transaction:

TX-48192

- Store:

Main Gift Shop

- Refund:

AED 1,850

- Cashier:

Employee 184

- Policy Threshold:

AED 500

- Status:

Approval Required

Actions:

Review | Approve | Reject | Request Information

- **RETURNS FRAUD / ANOMALY SUPPORT**

TICVAI should identify unusual return patterns without automatically accusing employees or guests.

Example:

Return rate for SKU-1842 is 3.8× above the category average.

Possible reasons:

Product Quality

Incorrect Sizing

Product Description

Packaging Damage

Fulfilment Error

Unusual Return Behavior

Actions:

Analyze Product | View Returns | View Guest Pattern | Review Supplier Quality

- **CROSS-STORE RETURNS**

The system should support configurable policies allowing products purchased from one store to be returned to another.

Example:

Purchased:

- **Waterpark Shop**

Return Requested:

- **Main Gift Shop**

TICVAI validates:

Same Tenant

Return Policy

Product Eligibility

Original Transaction

Tax

Tender

Inventory Destination

If allowed, the return proceeds without requiring the guest to return to the original location.

- **CROSS-CHANNEL RETURNS**

Where permitted, TICVAI should support:

- **Online Purchase → Store Return**
- **App Purchase → Store Exchange**
- **Store Purchase → Central Guest Service Refund**

The original transaction remains the single source of truth.

- **PRODUCT PERSONALIZATION FULFILMENT**

For personalized merchandise, the transaction shall create an operational fulfilment instruction.

Example:

- **Personalized Jersey**

- Name:

OMAR

- Number:

10

- Size:
M
- Base Price:
AED 120
- Personalization:
AED 35
- Total:
AED 155

Fulfilment:

- **Print Station 02**

Status:

- **In Production**

The workflow may be:

- **Paid**
 - → Personalization Queue
 - → In Production
 - → Quality Check
 - → Ready
 - → Guest Notified
 - → Collected
-

• **PHOTO / DIGITAL PRODUCT SUPPORT**

Retail shall support products associated with digital assets where applicable.

Example:

Guest scans:

- **Ticket / Wristband**
- ↓
- TICVAI retrieves eligible attraction photos
- ↓
- Guest selects photo
- ↓

Chooses:

- Digital Download
 - Printed Photo
 - Photo Frame
 - Photo Package
-

- ↓
- Payment
- ↓
- Digital / Physical Fulfilment

The actual media-management platform may be integrated externally, while Retail manages the commercial transaction.

• **INVENTORY COMMITMENT**

The Retail transaction shall communicate inventory events to the Shared Inventory Engine.

Example:

- **Physical POS Sale**
- **Sale Confirmed**
- → Inventory Consumption
- **Online Order**
- **Order Confirmed**
- → Inventory Reservation
- **Picking**
- → Inventory Allocated
- **Collection**
- → Inventory Issued
- **Return**
- → Inventory Inspection
- → Available / Damaged / Quarantine

Board 3 shall not maintain a separate Retail inventory ledger.

• **SERIALIZED / RFID PRODUCT SALE**

For serialized or individually identified products, the transaction shall capture the exact physical unit.

Example:

Product:

- Limited Edition Collectible
- SKU:
COL-001

Serial:

- **#184 / 500**

RFID EPC:
E280...

The sale shall associate that specific unit with the transaction.

Where configured, it may also be associated with the Guest Profile.

- **SHIFT & CASHIER INTEGRATION**

Retail POS shall integrate with TICVAI's common POS shift management.

The POS workspace should display:

- Current Cashier
- Shift Start
- Transaction Count
- Cash Drawer
- Tender Totals
- Exceptions
- Break Status

Cash-management functionality should use the shared POS/Finance architecture rather than creating a separate Retail cash-management engine.

- **EMPLOYEE APP INTEGRATION**

The TICVAI Employee App should support relevant Retail operational actions such as:

- View Orders
- Pick Order
- Pack Order
- Confirm Ready
- Scan Product
- Confirm Collection
- Process Approved Return
- View Stock Availability
- Request Stock
- Receive Alerts
- View Approval Requests

Example:

- **Click & Collect — Order R-18421**
 - 3 items
-

- SLA:
12 minutes

Action:

- **Start Picking**
-

• **AI-ASSISTED SELLING**

The POS may provide contextual recommendations to the cashier.

Example:

Guest cart:

- TICVAI T-Shirt

AI Suggestion:

72% of guests purchasing this T-shirt also purchase the TICVAI Cap.

Suggested:

- **Add Cap — AED 65**

AI recommendations should be configurable and should never obstruct high-speed checkout.

• **AI TRANSACTION RISK & ANOMALY DETECTION**

TICVAI should identify unusual operational patterns such as:

- Excessive Manual Discounts
- Repeated Voids
- High Refund Activity
- Unusual Price Overrides
- Abnormal Complimentary Items
- Excessive No-Receipt Returns
- Repeated Offline Transactions
- Unexpected Negative Sales

Example:

Manual discount activity at Exit Shop is 2.7× above its 30-day average.

TICVAI should surface the anomaly for review without automatically concluding fraud.

• **APPROVAL ENGINE INTEGRATION**

Sensitive Retail actions shall use TICVAI's common Approval Engine.

Examples:

- **High Discount**
 - → Store Manager
- **Price Override**
 - → Supervisor / Manager based on threshold
- **High-Value Refund**
 - → Retail Manager
- **No-Receipt Return**
 - → Authorized Role
- **Offline Exception**
 - → Operations
- **Inventory Write-Off**
 - → Inventory Manager

The workflow shall be configurable by tenant.

• **AUDIT TRAIL**

All sensitive Retail transaction actions shall be auditable.

Capture:

User
Role
Store
POS / Device
Date / Time
Transaction
Action
Original Value
New Value
Reason
Approval
Device
Online / Offline Status

Examples:

- **Discount Applied**
- **Price Overridden**
- **Item Voided**
- **Return Approved**
- **Refund Processed**

- **Transaction Reopened**
- **Offline Transaction Synchronized**

• **BOARD 3 — FINAL 10 SCREENS**

#	Screen
1	Retail Sales & POS Command Center
2	Retail POS Selling Workspace
3	Guest, Membership, Loyalty & Entitlement Interaction
4	Cart, Promotions, Discounts & Price Control
5	Checkout, Payment & Receipt
6	Transaction Search, Hold, Recall & Post-Sale Management
7	Returns, Refunds & Exchanges
8	Omnichannel Order & Fulfilment Center
9	Mobile POS, Event Sales & Offline Operations
10	Sales Exceptions, Controls & Operational Action Center

• **COMPLETE BOARD 3 TRANSACTION FLOW**

- **Store / Channel**
- ↓
- **Scan / Select Product**
- ↓
- **Product + SKU Validation**
- ↓
- **Check Availability**
- ↓
- **Identify Guest — Optional / Required by Policy**
- ↓
- **Resolve Price**
- ↓
- **Apply Membership / Loyalty / Promotion / Entitlement**
- ↓
- **Build Cart**
- ↓
- **Inventory Availability Check**
- ↓

- **Checkout**
- ↓
- **Payment Authorization**
- ↓
- **If Physical Sale**
- **Inventory Issue**
- ↓
- Receipt
- ↓
- Transaction Complete
- **If Omnichannel Order**
- **Inventory Reservation**
- ↓
- Fulfilment Location
- ↓
- Picking
- ↓
- Packing
- ↓
- Ready
- ↓
- Guest Notification
- ↓
- Collection / Delivery
- ↓
- Inventory Issue
- ↓
- Transaction Complete
- **If Return / Exchange**
- **Original Transaction Validation**
- ↓
- Return Policy
- ↓
- Product Condition
- ↓
- Approval if Required
- ↓
- Refund / Exchange / Credit

- ↓
 - Inventory Disposition
 - ↓
 - Finance + Loyalty Adjustment
 - ↓
 - Transaction Closed
-

- **ARCHITECTURAL REQUIREMENT**

Retail Board 3 shall use **one common TICVAI Commerce Transaction Engine** across all Retail channels.

Development should **not build separate transaction engines** for:

- **Fixed POS**
- **Mobile POS**
- **Kiosk**
- **B2C**
- **Guest App**
- **Click & Collect**
- **Event Sales**

Instead:

- **Commerce Transaction Engine**
- ↓
- Product / SKU
- ↓
- Price
- ↓
- Promotion
- ↓
- Guest / Loyalty / Entitlement
- ↓
- Inventory
- ↓
- Payment
- ↓
- Fulfilment
- ↓
- Return / Exchange

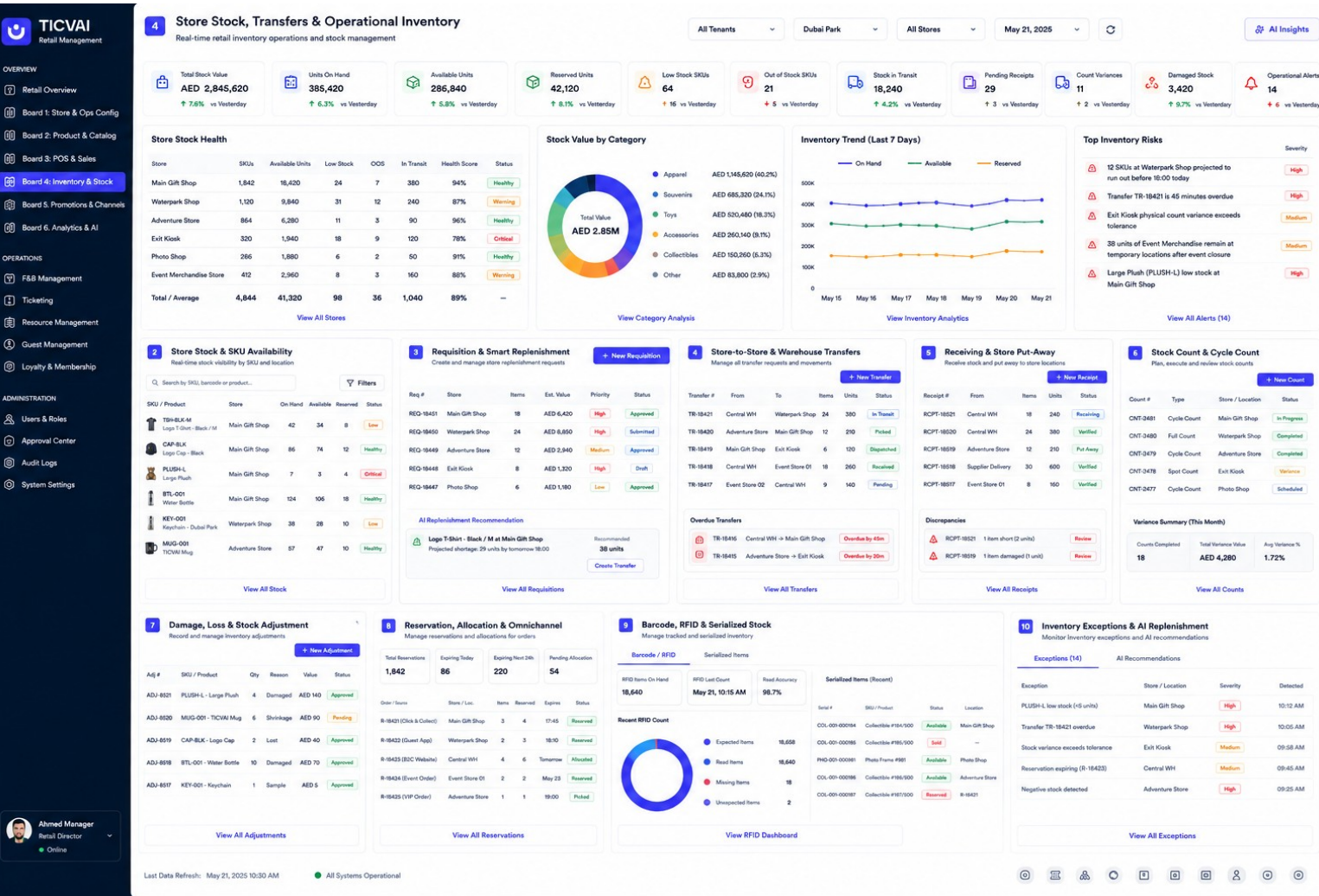
- ↓
- Finance
- ↓
- Analytics

The **channel changes the user experience, not the underlying commercial transaction logic.**

This is especially important for TICVAI because a guest may purchase a product through the mobile app, collect it from a physical store, exchange it at another store, receive loyalty points, and have all of those activities remain part of **one traceable transaction and guest journey.**

TICVAI RETAIL — BOARD 4

Store Stock, Transfers & Operational Inventory — Scope of Work



1. Purpose

The Store Stock, Transfers & Operational Inventory Board shall provide Retail teams with a real-time operational interface for managing stock availability, store replenishment, transfers, receiving, stock counts, reservations, damaged goods, serialized/RFID merchandise and inventory exceptions.

Board 4 shall serve as the **Retail operational layer** of TICVAI's Shared Inventory Engine.

It shall **not create a separate Retail inventory ledger**.

The architecture shall follow:

Retail Product / SKU → Inventory Item → Stock Location → Available / Reserved / In Transit / Damaged → Retail Operation → Shared Inventory Ledger

The Board shall support:

Retail Stores

Gift Shops

Kiosks

Mobile Carts

Pop-Up Stores

Event Merchandise Locations

Photo Shops

Online / Click & Collect Stock

Back-of-House Store Rooms

Central Warehouses

Temporary Event Stock Locations

The objective is to give Retail teams everything required to operate inventory while maintaining **one enterprise inventory source of truth across Retail, F&B and other TICVAI modules.**

PAGE 1 — Retail Inventory Command Center

Purpose

Provide Retail and Inventory managers with a centralized real-time view of stock health across all Retail locations.

Global Filters

- Tenant
- Company
- Park
- Venue
- Store
- Stock Location
- Warehouse
- Category
- Brand
- Collection
- Date

KPI Cards

- Total Retail Stock Value
- Units On Hand
- Available Units
- Reserved Units
- Low-Stock SKUs
- Out-of-Stock SKUs
- Overstocked SKUs
- Stock in Transit
- Pending Requisitions
- Pending Transfers
- Pending Receipts
- Count Variances
- Damaged Stock
- Stock Alerts

• Store Stock Health

Store	SKUs	Available	Low Stock	OOS	In Transit	Health
Main Gift Shop	1,842	18,420	24	7	380	94%
Waterpark Shop	1,120	9,840	31	12	240	87%

Store	SKUs	Available	Low Stock	OOS	In Transit	Health
Adventure Store	864	6,280	11	3	90	96%
Exit Kiosk	320	1,940	18	9	120	78%

• **Stock Risk Panel**

Examples:

12 SKUs at Waterpark Shop are projected to run out before 18:00.

Main Gift Shop has AED 18,400 of slow-moving Summer Collection stock.

Transfer TR-18421 is 45 minutes overdue.

Exit Kiosk physical count variance exceeds tolerance.

38 units of Event Merchandise remain at temporary locations after event closure.

Quick Actions

Stock Lookup | Create Requisition | Request Transfer | Receive Stock | Start Count | Record Damage | AI Replenishment

PAGE 2 — Store Stock & SKU Availability

Purpose

Provide detailed stock visibility for every Retail SKU across stores, warehouses and other approved stock locations.

- **Inventory Status Dimensions**

For each SKU, display:

On Hand
Available
Reserved
Allocated
Picked
In Transit
Incoming
Damaged
Quarantine
Blocked
Returned
Available-to-Promise

- **Stock Table**

SKU	Product	On Hand	Reserved	Available	Par	Incoming	Status
TSH-BLK-M	Logo T-Shirt	42	8	34	60	20	Low
CAP-BLK	Logo Cap	86	12	74	60	0	Healthy
PLUSH-L	Large Plush	7	4	3	20	0	Critical
BTL-001	Water Bottle	124	18	106	80	40	Healthy

- **SKU Drill-Down**

Selecting a SKU shall display:

Product
Variant
Barcode
RFID Status
Current Location
Bin / Shelf
On Hand
Available

Reserved

Incoming

In Transit

Damaged

Last Receipt

Last Sale

Last Transfer

Last Count

Current Cost

Stock Value

Replenishment Settings

- **Multi-Location Stock Lookup**

Example:

- **TICVAI Logo T-Shirt — Black / Medium**

- Main Gift Shop:
34 available
- Waterpark Shop:
4 available
- Adventure Store:
18 available
- Exit Kiosk:
0 available
- Central Warehouse:
220 available
- Event Warehouse:
68 available

This information shall allow TICVAI to recommend internal stock redistribution before procurement.

PAGE 3 — Requisition & Smart Store Replenishment

Purpose

Allow Retail stores to request stock from warehouses or other authorized supply locations and enable automated replenishment recommendations.

- **Requisition Information**

Capture:

Requisition Number

Requesting Store

Supply Location

Required Date

Required Time

Priority

Product

SKU

Current Stock

Available Stock

Par Level

Minimum Level

Maximum Level

Forecast Demand

Reserved Quantity

Incoming Quantity

Requested Quantity

Recommended Quantity

UOM

Reason

- **Example**

- **Logo T-Shirt — Black / Medium**

- Available:

12

- Minimum:

15

- Par:

50

- Expected Sales Today:

34

- Incoming:
0

Recommended Replenishment:

- **38 Units**
- **Requisition Lifecycle**
- **Draft**
- → Submitted
- → Approval
- → Approved
- → Warehouse Picking
- → Packed
- → Dispatched
- → In Transit
- → Received
- → Closed
- **Replenishment Modes**

Support:

- Manual
- Scheduled
- Min / Max
- Par Stock
- Sales Velocity
- Demand-Based
- Event-Based
- Forecast-Based
- AI Recommended
- **Auto-Requisition**

Where enabled:

- **Stock / Forecast Trigger**
- ↓
- TICVAI creates proposed requisition
- ↓
- Authorized user reviews
- ↓
- Approve / Modify / Reject

The system shall not automatically create external purchase orders from Retail.

If internal stock is unavailable, the requirement shall be passed to the Shared Procurement workflow.

PAGE 4 — Store-to-Store & Warehouse Transfers

Purpose

Manage stock movement between Retail stores, warehouses and temporary selling locations.

- **Supported Transfer Types**

Warehouse → Store

Store → Store

Store → Warehouse

Warehouse → Event Store

Event Store → Warehouse

Store → Pop-Up Location

Temporary Location → Store

Store → Fulfilment Location

- **Transfer Request**

Capture:

Transfer Number

Source

Destination

Required Date

Priority

SKU

Requested Quantity

Approved Quantity

Picked Quantity

Dispatched Quantity

Received Quantity

Reason

Requester

- **Transfer Lifecycle**

- **Requested**

- → Approved

- → Picking

- → Packed

- → Dispatched

- → In Transit

- → Partially Received / Received

- → Reconciled

- → Closed
- **Transfer Example**
- **TR-18421**

From:

- Central Warehouse

To:

- Waterpark Shop

Items:

- 24 SKUs

Units:

- 380

Status:

- **In Transit**

ETA:

- 11:45
- **Emergency Transfer**

Example:

Large Plush projected to stock out at Main Gift Shop in 45 minutes.

Nearby availability:

- Adventure Store:
38
- Central Warehouse:
120

TICVAI Recommendation:

Transfer 20 units from Adventure Store immediately.

Actions:

Create Transfer | Select Different Source | Ignore

PAGE 5 — Receiving & Store Put-Away

Purpose

Provide Retail stores with controlled receiving of stock transferred from warehouses, other stores or approved supply processes.

- **Receiving Methods**

Support:

Transfer Receipt

Warehouse Delivery

Procurement Delivery where authorized

Event Stock Receipt

Return to Store

Inter-Store Receipt

- **Receipt Header**

Capture:

Receipt Number

Source

Destination

Transfer / PO Reference

Dispatch Date

Receipt Date

Receiver

Shipment / Container Reference

Status

- **Receiving Verification**

Employees should be able to verify items using:

Barcode

QR

RFID

Manual Search

- **Example**

Expected:

- Logo T-Shirt / M

- 50 units

Received:

- 48

Damaged:

- 1

Missing:

- 1

TICVAI creates:

- **Receiving Discrepancy — 2 units**
- **Discrepancy Types**

Short

Over

Damaged

Wrong Product

Wrong Variant

Missing

Unexpected Item

Serial Mismatch

RFID Mismatch

- **Put-Away**

After receiving, employees may assign stock to:

Store Room

Shelf

Bin

Display

Pickup Area

Secure Storage

- **Receipt Lifecycle**

- **Expected**

- → Receiving
 - → Verified
 - → Discrepancy Review where applicable
 - → Put Away
 - → Available
-
-

PAGE 6 — Stock Count & Cycle Count Management

Purpose

Support accurate physical stock counting across Retail locations.

- **Count Types**

Opening Count

Closing Count

Shift Count

Spot Count

Cycle Count

Category Count

Full Store Count

Event Close Count

Investigation Count

- **Count Creation**

Configure:

Store

Location

Category

Product Range

Count Type

Scheduled Date

Assigned Employees

Blind Count

Recount Rules

Variance Tolerance

- **Blind Count**

When enabled:

Employees shall **not see expected system quantities** while counting.

- **Count Methods**

Support:

Barcode Scan

RFID Scan

Manual Quantity

Mobile Device

Employee App

- **Count Results**

SKU	System	Physical	Variance	Variance %	Value
TSH-BLK-M	42	40	-2	-4.8%	-AED 90
CAP-BLK	86	86	0	0%	AED 0
PLUSH-L	7	5	-2	-28.6%	-AED 70

- **Variance Workflow**

- **Count Completed**

- ↓
- Variance Detected
- ↓
- Tolerance Check
- ↓
- Within Policy
- → Process according to configuration
- Outside Policy
- → Recount / Investigation
- ↓
- Approval
- ↓
- Shared Inventory Adjustment

- **Count Locking**

Where required, the system shall support temporary transaction controls while a full physical count is being conducted.

PAGE 7 — Damage, Loss, Shrinkage & Stock Adjustment

Purpose

Provide controlled management of inventory that cannot remain in normal sellable stock.

- **Adjustment Reasons**

Configurable examples:

- Damaged
- Defective
- Lost
- Theft / Shrinkage
- Display Damage
- Packaging Damage
- Sample
- Promotional Use
- Employee Use
- Event Loss
- Count Correction
- System Correction
- Other

- **Stock Adjustment**

Capture:

- Store
- SKU
- Quantity
- Current Status
- New Status
- Reason
- Cost
- Estimated Retail Value
- Employee
- Date / Time
- Notes
- Photo Evidence, optional

- **Example**

- **Limited Edition Mug**

- Quantity:
6

- Reason:
Damaged During Display
- Cost:
AED 108
- Retail Value:
AED 270

Action:

- **Move to Damaged Stock**
- **Approval Threshold**

Example:

- Up to AED 250
- → Store Supervisor
- AED 251–1,000
- → Store Manager
- Above AED 1,000
- → Retail / Inventory Manager

Thresholds shall be configurable.

- **Shrinkage Analysis**

TICVAI shall track:

- **Expected Stock**
- vs
- **Physical Stock**
- vs
- **Recorded Adjustments**

to support shrinkage analytics in Board 6.

The system shall surface discrepancies for investigation without automatically attributing them to theft or employee misconduct.

PAGE 8 — Reservation, Allocation & Omnichannel Inventory

Purpose

Manage Retail inventory commitments created by digital and omnichannel orders.

This page is particularly important because **On Hand Stock is not the same as Available Stock.**

- **Inventory States**

Example:

- On Hand:
100
- Reserved:
20
- Allocated:
10
- Picked:
5

Available:

- **65**
- **Reservation Sources**

Inventory may be reserved for:

B2C Order
Guest App Order
Click & Collect
Event Order
VIP Order
Personalized Product
Store Pickup
Delivery Order

- **Reservation Lifecycle**

- **Available**

- ↓
- Reserved
- ↓
- Allocated
- ↓
- Picked
- ↓
- Packed

- ↓
- Collected / Delivered
- ↓
- Issued
- **Reservation Expiry**

Example:

Click & Collect Order:

- R-18421

Reservation:

- 4 units

Hold Until:

- 18:00

If the guest does not collect within the configured period:

- **Reservation Expired**
- ↓
- Stock Released

unless another policy applies.

- **Available-to-Promise**

TICVAI should calculate whether an item can be promised based on:

Current Available

Existing Reservations

Incoming Transfers

Expected Receipt

Fulfilment SLA

Store / Warehouse Eligibility

- **Cross-Location Fulfilment**

Example:

Online order requests:

- Limited Edition Plush
- Main Gift Shop:
0
- Waterpark Shop:
2
- Central Warehouse:
40

TICVAI may recommend:

Fulfil from Central Warehouse rather than rejecting the order.

PAGE 9 — Barcode, RFID, Serialized Stock & Traceability

Purpose

Provide advanced operational management for merchandise requiring individual identification or rapid stock scanning.

- **Barcode Operations**

Support barcode use for:

- Receiving
- Selling
- Stock Lookup
- Transfer
- Picking
- Packing
- Stock Count
- Return
- Adjustment

- **RFID Operations**

Where enabled:

- Bulk Stock Count
- Receiving
- Transfer Verification
- Store Movement
- Item Search
- Exit Verification
- Inventory Audit

- **RFID Count Example**

Expected Items:

- 1,842

RFID Read:

- 1,836

Missing:

- 6

Unexpected:

- 2

Actions:

Review Missing | Review Unexpected | Re-Scan

- **Serialized Products**

For individually tracked products:

Serial Number

SKU

Barcode

RFID EPC

Current Location

Inventory Status

Receipt

Transfer History

Sale

Return

Guest Link where applicable

- **Example**

- **Limited Edition Collectible**

- SKU:

COL-001

- Serial:

#184 / 500

- Status:

Available

- Location:

Main Gift Shop / Secure Cabinet 02

- **Traceability Timeline**

- **Received**

- ↓

- Warehouse

- ↓

- Transferred

- ↓

- Main Gift Shop

- ↓

- Moved to Display

- ↓

- Sold

- ↓

- Returned

- ↓

- Inspected
- ↓
- Available / Quarantine

This complete history shall remain accessible according to role permissions.

PAGE 10 — Inventory Exceptions, AI Replenishment & Action Center

Purpose

Provide a centralized operational workspace for Retail inventory risks, exceptions and AI recommendations.

- **Alert Categories**

- **Stock**

Low Stock

Critical Stock

Out of Stock

Overstock

Negative Stock

- **Replenishment**

Requisition Pending

Replenishment Late

Insufficient Source Stock

- **Transfers**

Transfer Delayed

Partial Transfer

Receiving Discrepancy

Missing Transfer

- **Counts**

Count Variance

Recount Required

Unapproved Adjustment

- **Omnichannel**

Reservation Conflict

Fulfilment Stock Shortage

Reservation Expiring

- **Devices**

RFID Reader Offline

Scanner Offline

Sync Failure

- **Severity**

Information | Warning | High | Critical

- **AI — SMART REPLENISHMENT**

TICVAI should not depend solely on:

- **Stock < Minimum → Reorder**

The recommendation engine should consider:

Current Available Stock

Reserved Stock

Incoming Stock

Sales Velocity

Historical Demand

Current Attendance

Future Ticket Sales

Events

Seasonality

Promotions

Product Lifecycle

Store Type

Safety Stock

Internal Lead Time

Supplier Lead Time

- **Example**

- **Logo T-Shirt — Medium**

- Main Gift Shop

- Available:

22

- Tomorrow Forecast:

48

- Reserved:

5

- Incoming:

10

- Safety Stock:

8

Projected Shortfall:

- **29 Units**

Recommendation:

Transfer 30 units from Central Warehouse before tomorrow's opening.

Actions:

Simulate | Create Transfer | Create Requisition | Ignore

- **AI — INTERNAL STOCK REDISTRIBUTION**

Before recommending procurement, TICVAI should evaluate stock across the network.

Example:

- **Limited Edition Cap**

Main Gift Shop:

- Projected shortage:
40

Waterpark Shop:

- Projected excess:
22

Adventure Store:

- Projected excess:
28

AI Recommendation:

- **Waterpark → Main Gift Shop: 20**
- **Adventure → Main Gift Shop: 20**

External Procurement:

- **Not Required**

This reduces unnecessary purchasing while improving product availability.

- **AI — EVENT STOCK PLANNING**

TICVAI should use ticketing and event data to predict merchandise requirements.

Example:

- **Concert — Saturday**

Expected Attendance:

- 18,500

Historical Merchandise Conversion:

- 21%

Expected Retail Transactions:

- 3,885

Expected Top Product:

- Concert T-Shirt

Forecast Demand:

- 2,150 units

Current Event Stock:

- 1,420

Incoming:

- 400

Projected Shortfall:

- **330 units**

Recommendation:

Allocate an additional 350 units before event opening.

This integration between **Ticketing + Events + Retail Inventory** should be a key TICVAI differentiator.

• **AI — SLOW-MOVING & OVERSTOCK IDENTIFICATION**

TICVAI shall identify stock where supply materially exceeds expected demand.

Example:

- **Summer Plush Collection**

Stock Value:

- AED 84,000

30-Day Sales:

- AED 12,400

Projected Days of Supply:

- **184 Days**

AI Recommendation:

Reduce future replenishment

Transfer stock to higher-demand stores

Consider promotion

Consider bundle

Review markdown scenario

Actual pricing/promotion changes shall be handled through the appropriate Retail boards and Approval Engine.

• **AI — STOCK-OUT LOST SALES**

TICVAI should estimate commercial impact when a Retail product becomes unavailable.

Example:

- **Limited Edition Plush**

Out of Stock:

- 14:10–18:35

Expected Units:

- 68

Actual:

- 4

Estimated Lost Sales:

- **AED 6,080**

Estimated Lost Margin:

- **AED 3,420**

Recommended Action:

Increase weekend safety stock from 15 to 45 units.

This information shall feed Board 6 Retail Intelligence.

- **AI — INVENTORY ANOMALY DETECTION**

The system should identify abnormal patterns such as:

Unexpected Stock Variance

Repeated Negative Stock

Excessive Adjustments

Unusual Damage

Repeated Transfer Discrepancies

High Return-to-Stock Variance

RFID / Physical Count Difference

Example:

Waterpark Shop stock variance for souvenir mugs is 4.2× above the 30-day average.

Potential areas to investigate:

Receiving

Transfer

POS Posting

Returns

Damage

Count Accuracy

TICVAI shall not automatically conclude fraud or misconduct.

- **EMPLOYEE APP INTEGRATION**

Board 4 shall integrate directly with the TICVAI Employee App.

Authorized Retail employees should be able to:

- Scan SKU
- View Stock
- Check Other Locations
- Request Replenishment
- Request Transfer
- Pick Transfer
- Dispatch Transfer
- Receive Transfer
- Count Stock
- Record Damage
- Process Put-Away
- Pick Online Order
- Confirm Packing
- Locate RFID Item
- View Alerts

- **Example**

- **Low Stock — Large Plush**

- Available:
3
- Expected Demand:
18

Nearby Stock:

- Adventure Store:
28

Action:

- **Request Transfer**
-

- **APPROVAL ENGINE INTEGRATION**

Board 4 shall use TICVAI's shared Approval Engine.

Examples:

- **Stock Adjustment**
 - → Store Manager
- **High-Value Write-Off**
 - → Retail Manager + Inventory
- **Emergency Transfer**
 - → Configurable Approval
- **Count Variance**
 - → Inventory Supervisor
- **Requisition**
 - → Store Manager / Inventory Manager
- **Negative Stock Override**
 - → Authorized Inventory Role

No separate Retail approval engine shall be created.

• **SHARED INVENTORY & PROCUREMENT BOUNDARY**

This separation is mandatory.

• **RETAIL BOARD 4 SHALL OWN**

Store Stock Visibility

SKU Availability

Retail Replenishment Requests

Store Transfer Requests

Retail Receiving Interface

Store Put-Away

Retail Stock Counts

Retail Damage / Loss Recording

Retail Reservations

Omnichannel Stock Allocation

RFID / Barcode Retail Operations

Retail Stock Alerts

Retail AI Replenishment Recommendations

• **SHARED INVENTORY SHALL OWN**

Enterprise Inventory Item Master

Stock Ledger

Warehouse Master

Storage Locations / Bins

Enterprise Stock Transactions

Transfer Posting

Goods Receipt

Inventory Adjustments

Inventory Valuation

Costing Methods

Batch / Serial Master

Enterprise Inventory Reconciliation

- **PROCUREMENT SHALL OWN**

Supplier Master

Supplier Contracts

Purchase Requisitions

RFQs

Quotations

Quotation Comparison

Purchase Orders

Supplier Deliveries

Supplier Invoices

Supplier Performance

- **RETAIL REPLENISHMENT FLOW**

- **Retail Demand**

- ↓

- Current Store Availability

- ↓

- Reservations

- ↓

- Forecast Demand

- ↓

- Minimum / Par / Safety Stock

- ↓

- **Replenishment Requirement**

- ↓

- Check Internal Stock

- **If Available Internally**

- **Create Transfer**

- ↓

- Warehouse / Store Picking
- ↓
- Dispatch
- ↓
- Receive
- ↓
- Put Away
- ↓
- Store Stock Available
- **If Internal Stock Is Insufficient**
- **Create Requisition**
- ↓
- Shared Procurement
- ↓
- Purchase Order
- ↓
- Supplier Delivery
- ↓
- Shared Inventory Receipt
- ↓
- Warehouse Stock
- ↓
- Retail Replenishment
- ↓
- Store Availability Updated

• BOARD 4 — FINAL 10 SCREENS

#	Screen
1	Retail Inventory Command Center
2	Store Stock & SKU Availability
3	Requisition & Smart Store Replenishment
4	Store-to-Store & Warehouse Transfers
5	Receiving & Store Put-Away
6	Stock Count & Cycle Count Management

#	Screen
7	Damage, Loss, Shrinkage & Stock Adjustment
8	Reservation, Allocation & Omnichannel Inventory
9	Barcode, RFID, Serialized Stock & Traceability
10	Inventory Exceptions, AI Replenishment & Action Center

- **COMPLETE BOARD 4 OPERATIONAL FLOW**

- **Retail Sale / Online Order / Forecast**

- ↓

- **SKU Demand**

- ↓

Check:

- **On Hand – Reserved – Allocated = Available**

- ↓

- **If Stock Is Healthy**

- Continue Selling

- **If Stock Is Low**

- TICVAI calculates replenishment requirement

- ↓

- Check Internal Network

- ↓

- **Internal Stock Available**

Recommend:

- **Warehouse Transfer / Store-to-Store Transfer**

- ↓

- Approval where required

- ↓

- Pick

- ↓

- Dispatch

- ↓

- Receive

- ↓

- Put Away

- ↓

- Available Stock Updated

- **Internal Stock Not Available**

Create:

- **Requisition**
- ↓
- Shared Procurement
- ↓
- Goods Received into Shared Inventory
- ↓
- Retail Replenishment
- ↓
- Stock Available

Meanwhile:

POS + B2C + Guest App + Click & Collect

consume the same real-time availability.

- ↓
- Physical Count
- ↓
- Variance Detection
- ↓
- Investigation / Approval
- ↓
- Shared Inventory Adjustment
- ↓

Board 6 Analytics & AI.

• **ARCHITECTURAL REQUIREMENT**

The most important requirement for Board 4 is that Development must **not build a Retail Inventory System separate from the Inventory & Procurement module we will design later.**

There should be **one TICVAI Shared Inventory Engine:**

- **Inventory Item**
- → Location
- → Quantity
- → Status
- → Reservation
- → Movement
- → Cost
- → Transaction
- → Audit

Then each operational module uses that engine differently:

Retail

SKU + Store + Sale + Return + Omnichannel Reservation

F&B

Ingredient + Recipe + Production + Waste + 86

- **Rental**

Individual Asset + Check-Out + Check-In

- **Operations**

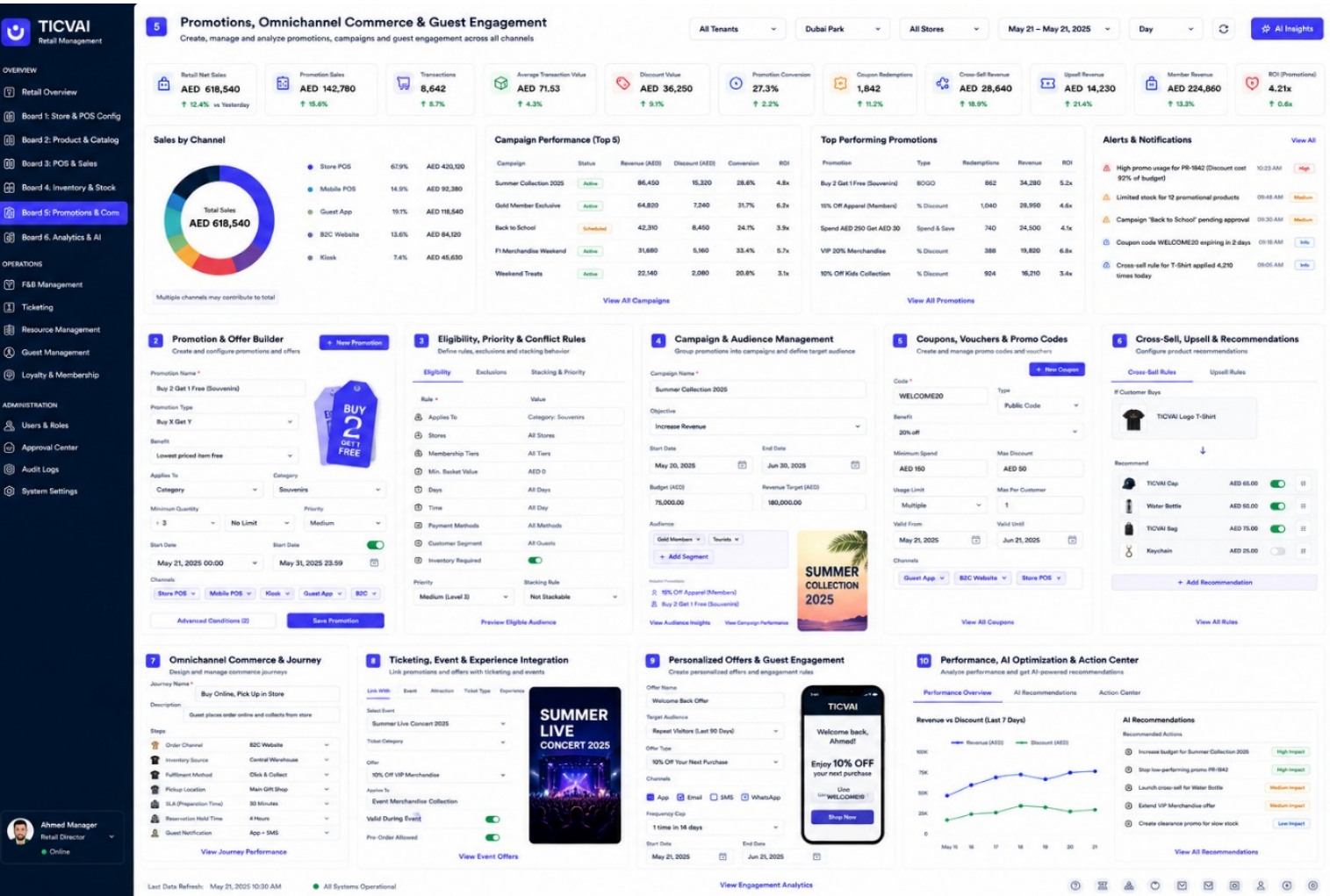
Consumables + Equipment + Internal Usage

This is particularly important because when we start the **Shared Inventory & Procurement boards**, we should not redesign the Retail stock functionality again. Instead, those boards will provide the enterprise engine underneath both **F&B Board 5 and Retail Board 4**.

That architecture will give TICVAI **one stock source of truth across the entire platform**, while still giving Retail users a backend designed specifically for their day-to-day operational needs.

TICVAI RETAIL — BOARD 5

Promotions, Omnichannel Commerce & Guest Engagement — Scope of Work



1. Purpose

The Promotions, Omnichannel Commerce & Guest Engagement Board shall provide the centralized commercial orchestration layer for Retail, allowing TICVAI operators to configure **promotions, campaigns, coupons, vouchers, cross-sell, upsell, omnichannel journeys, guest targeting and personalized retail experiences** across physical and digital sales channels.

The Board shall answer five key commercial questions:

What offer should we provide? → To whom? → Where? → When? → Through which channel?

The architecture shall follow:

Guest / Segment → Product / SKU → Offer → Eligibility → Channel → Store → Inventory → Transaction → Fulfilment → Measurement

Board 5 shall work across:

- Fixed POS
- Mobile POS

Self-Service Kiosk

B2C Website

Guest Mobile App

QR Commerce

Click & Collect

Event Merchandise

Retail Stores

Pop-Up Stores

Other integrated commerce channels

The Board shall integrate with TICVAI's shared:

Product & Catalog Engine

Pricing Engine

Promotion Engine

Guest Profile / CRM

Loyalty & Membership

Wallet / Entitlement Engine

Inventory Engine

Payment Engine

Ticketing & Attendance

Events & Experiences

Approval Engine

Finance

Analytics & AI

The objective is to avoid building promotions separately inside POS, mobile, B2C or individual stores.

There shall be **one TICVAI Promotion & Commerce Rules Engine** consumed by all channels.

PAGE 1 — Retail Commercial Command Center

Purpose

Provide Retail and Commercial management with a consolidated real-time view of promotions, campaigns, omnichannel sales and guest engagement.

Global Filters

Tenant
Company
Park
Venue
Store
Channel
Campaign
Promotion
Guest Segment
Membership Tier
Date / Period

KPI Cards

Retail Net Sales
Promotion Revenue
Promotion Transactions
Discount Value
Average Transaction Value
Units per Transaction
Promotion Conversion
Coupon Redemptions
Cross-Sell Revenue
Upsell Revenue
Digital Retail Revenue
Click & Collect Orders
Member Revenue
Campaign ROI

• **Channel Performance**

Example:

Channel	Sales	Transactions	ATV	Promotion Use	Growth
Store POS	AED 420K	6,820	AED 61.6	24%	+6%

Channel	Sales	Transactions	ATV	Promotion Use	Growth
Mobile POS	AED 92K	1,180	AED 78.0	18%	+14%
Guest App	AED 118K	1,420	AED 83.1	31%	+22%
B2C	AED 84K	940	AED 89.4	28%	+18%
Kiosk	AED 46K	720	AED 63.9	16%	+9%

• Campaign Snapshot

Display:

Campaign

Status

- Audience
- Revenue
- Discount Cost
- Conversion
- ROI
- End Date

• Commercial Alerts

Examples:

- Summer Merchandise campaign is 18% above revenue target.
- Member promotion redemption is below forecast.
- Promotion PR-1842 is producing margin below configured threshold.
- 14 campaign products are projected to stock out today.
- Online conversion increased 21% after cross-sell activation.

Quick Actions

+ Promotion | + Campaign | + Coupon | + Segment Offer | Cross-Sell | View Performance

PAGE 2 — Promotion & Offer Builder

Purpose

Allow authorized commercial users to configure Retail promotions without software development.

- **Promotion Types**

The system shall support configurable promotion structures including:

- Percentage Discount
- Fixed Amount Discount
- Fixed Promotional Price
- Buy X Get Y
- Buy One Get One
- Multi-Buy
- Bundle Price
- Spend Threshold
- Quantity Discount
- Category Discount
- Product Discount
- Member Discount
- VIP Offer
- Coupon
- Voucher
- Event Offer
- Time-Based Offer
- Store-Specific Offer
- Channel-Specific Offer

- **Example**

- **Buy 2 Get 1 Free**

Eligible Category:

- Souvenirs

Minimum Quantity:

- 3

Benefit:

- Lowest-Priced Eligible Item Free

Stores:

- All Dubai Park Retail Stores

Channels:

- POS + Mobile + App

Effective:

- 1 Sep – 30 Sep
- **Promotion Builder**

The interface should follow:

- **IF**

Guest / Product / Store / Channel / Date / Quantity / Spend condition

- **THEN**

Apply Discount / Free Product / Fixed Price / Voucher / Benefit

- **Example Rule**

- **IF**

- Guest Membership = Gold
- AND
- Category = Apparel
- AND
- Spend $\geq$ AED 200

- **THEN**

- Apply 15% Discount

- **Rule Groups**

Support:

- **AND / OR**

logic for advanced eligibility.

PAGE 3 — Promotion Eligibility, Priority & Conflict Rules

Purpose

Control which offers may be applied and resolve situations where several promotions are simultaneously eligible.

- **Eligibility Dimensions**

Promotions may be restricted by:

- Product
- SKU
- Category
- Brand
- Collection
- Store
- Venue
- Park
- Channel
- Guest
- Guest Segment
- Membership Tier
- Ticket Type
- Event
- Date
- Time
- Day of Week
- Quantity
- Basket Value
- Payment Method
- Inventory Availability

- **Exclusions**

Allow:

- Excluded Products
- Excluded Categories
- Excluded Stores
- Excluded Channels
- Excluded Guests
- Excluded Dates
- Non-Discountable Products

- **Promotion Priority**

Example:

Contractual / Entitlement Benefit

VIP Benefit

Membership Price

Campaign Promotion

Coupon

Manual Discount

Priority shall be tenant-configurable.

- **Stacking Rules**

Support:

Stackable

Non-Stackable

Exclusive

Best Price

Highest Priority

Maximum Discount Cap

Customer Choice

- **Conflict Example**

Guest qualifies for:

- Gold Member:
10%
- Summer Campaign:
15%
- Coupon:
AED 20

TICVAI calculates permitted combinations and displays:

- **Best Eligible Benefit: AED 42 Saving**
- **Margin Protection**

Promotion execution shall be capable of checking:

Product Cost

Current Price

Promotion Price

Expected Margin

Minimum Margin Threshold

Example:

- Promotion Price:
AED 70
- Cost:
AED 48
- Margin:
31.4%
- Required Minimum:
40%

Result:

- **Approval Required**
-
-

PAGE 4 — Campaign & Audience Management

Purpose

Allow Retail teams to group offers into structured commercial campaigns and define the intended audience.

- **Campaign Information**

Capture:

Campaign Name

Campaign Code

Description

Objective

Start Date

End Date

Budget

Revenue Target

Target Audience

Stores

Channels

Products / Collections

Promotions

Status

- **Campaign Objectives**

Examples:

Increase Revenue

Increase ATV

Increase Units per Transaction

Clear Slow-Moving Stock

Launch New Product

Increase Member Spend

Increase Digital Conversion

Drive Event Merchandise Sales

Increase Cross-Sell

Recover Abandoned Cart

- **Audience Segments**

Campaigns may use guest segments such as:

All Guests

Members

VIP

Families

Tourists

Residents

Repeat Visitors

High-Spend Guests

Event Attendees

Ticket Holders

Specific Membership Tier

Custom CRM Segment

Guest data shall come from the shared Guest / CRM layer.

- **Dynamic Segment Example**

Guests who:

- Visited Dubai Park
- within last 90 days
- AND
- Purchased Retail $\geq$ AED 300
- AND
- Did not purchase Apparel

- **Campaign Lifecycle**

- **Draft**

- → Review
 - → Approval
 - → Scheduled
 - → Active
 - → Paused
 - → Completed
 - → Performance Review
-
-

PAGE 5 — Coupons, Vouchers & Promotional Codes

Purpose

Provide centralized management of Retail coupons and promotional codes.

- **Coupon Types**

Support:

- Single Use
- Multi-Use
- Guest-Specific
- Public Code
- Campaign Code
- QR Coupon
- Barcode Coupon
- Ticket-Linked Coupon
- Membership Coupon
- Event Coupon

- **Coupon Configuration**

Capture:

- Coupon Name
- Code
- Barcode / QR
- Promotion
- Value
- Valid From
- Valid Until
- Maximum Redemptions
- Maximum per Guest
- Eligible Products
- Eligible Stores
- Eligible Channels
- Guest Segment
- Minimum Spend

- **Example**

- **WELCOME20**

Benefit:

- 20% Off

Minimum Spend:

- AED 150

Maximum Discount:

- AED 50

Usage:

- 1 per Guest

Channels:

- Guest App + B2C
- **Distribution**

Coupons may be distributed through:

Email

SMS

WhatsApp where integrated

Guest App

Ticket Confirmation

Membership

QR Campaign

Physical Voucher

- **Redemption Tracking**

Display:

- Issued:
25,000
 - Activated:
8,420
 - Redeemed:
5,840
 - Conversion:
23.4%
 - Revenue Generated:
AED 486K
-
-

PAGE 6 — Cross-Sell, Upsell & Recommendation Rules

Purpose

Increase Retail basket value by recommending relevant complementary or upgraded products.

- **Cross-Sell**

Example:

Guest selects:

- **TICVAI T-Shirt**

Recommend:

Cap

Water Bottle

Bag

- **Upsell**

Example:

Guest selects:

- Standard Plush — AED 65

Recommend:

- **Limited Edition Plush — AED 95**

- **Recommendation Sources**

Recommendations may be:

Manually Configured

Product Relationship

Category Based

Basket Based

Guest Segment Based

Historical Purchase Based

AI Recommended

- **Rule Example**

- **IF**

- Cart contains:

T-Shirt

- AND

- No Cap in Cart

- **THEN**

- Recommend:

Logo Cap

- **Channel Placement**

Recommendations may appear on:

POS

Mobile POS

Kiosk

B2C

Guest App

Cart

Checkout

Order Confirmation

- **Recommendation Limits**

Configure:

Maximum Recommendations

Priority

Stock Requirement

Margin Requirement

Channel

Guest Eligibility

Products with insufficient inventory should not normally be promoted unless explicitly permitted.

PAGE 7 — Omnichannel Commerce & Journey Configuration

Purpose

Configure consistent Retail customer journeys across physical and digital channels.

- **Supported Journeys**

Examples:

- **Browse Online → Buy Online → Collect in Store**
- **Browse App → Reserve Product → Buy at Store**
- **Buy at Store → Deliver to Hotel**
- **Buy Online → Return at Store**
- **Buy at Waterpark → Exchange at Main Gift Shop**
- **Event Ticket → Pre-Order Merchandise → Collect at Event**
- **Journey Builder**

Configure:

- **Source Channel**
- ↓
- **Transaction Channel**
- ↓
- **Inventory Source**
- ↓
- **Fulfilment Method**
- ↓
- **Collection / Delivery Location**
- ↓
- **Return Location**
- **Example**
- **Click & Collect**

Order Channel:

- Guest App

Inventory:

- Central Warehouse

Collection:

- Main Gift Shop

Preparation SLA:

- 30 minutes

Reservation Hold:

- 4 hours

Guest Notification:

- App + SMS
- **Cross-Channel Rules**

Configure:

Cross-Store Return

Cross-Channel Return

Cross-Store Exchange

Online-to-Store Return

Store-to-Delivery

Reserve Online / Pay Store

Buy Online / Collect Event

- **Journey Availability**

Journeys may be enabled by:

Tenant

Park

Venue

Store

Product

Channel

Guest Type

PAGE 8 — Ticketing, Event & Experience Commerce Integration

Purpose

Connect Retail commerce directly with TICVAI's core Ticketing, Event and Experience platform.

This should be a significant differentiator for TICVAI.

- **Ticket + Merchandise**

Example:

Guest purchases:

- **Theme Park Admission**

TICVAI offers:

Add Limited Edition Park T-Shirt for AED 75 and collect at Main Gift Shop.

- **Event Merchandise**

Event:

- Concert 2026
- Ticket Purchased:
VIP

TICVAI may offer:

Event T-Shirt
Cap
Poster
VIP Collectible
Pre-Order Package

- **Ticket-Based Eligibility**

Promotions may depend on:

Ticket Product
Ticket Category
Event
Attraction
Membership
Package
Visit Date
VIP Status

- **Example**

- **IF**

- Guest holds VIP Concert Ticket

- **THEN**

- Enable exclusive merchandise catalog
- AND
- 10% VIP Merchandise Benefit
- **Attendance-Based Offer**

Example:

Guest scans into Waterpark.

At 15:00:

Enjoy 15% off Waterpark merchandise before 18:00.

- **Attraction-Based Merchandise**

Where appropriate, TICVAI may associate merchandise with:

Attraction

Experience

Event

Tour

Venue

Example:

- Guest completes attraction
- ↓

Relevant photo / souvenir / collectible offer becomes available.

PAGE 9 — Personalized Offers & Guest Engagement

Purpose

Allow TICVAI to deliver contextual Retail offers using guest behavior, transaction history, membership, visit context and available inventory.

- **Personalization Inputs**

Where permitted and available:

- Guest Purchase History
- Visit History
- Membership
- Loyalty
- Ticket / Event
- Previous Retail Categories
- Basket Behavior
- Current Location / Venue Context
- Language
- Preferred Channel
- Campaign Response

- **Example**

Guest:

- Gold Member

Previous Purchases:

- Kids Apparel + Plush

Current Visit:

- Waterpark

TICVAI Recommendation:

New Waterpark Kids Collection available today — Gold Members receive 10% off.

- **Next-Best-Offer**

TICVAI may rank eligible offers based on:

- Relevance
- Probability of Purchase
- Margin
- Stock Availability
- Campaign Priority
- Guest Eligibility

- **Communication Channels**

Where consent and integration permit:

Guest App

Email

SMS

WhatsApp

B2C

POS

Kiosk

- **Frequency Controls**

Configure:

Maximum Offers per Day

Contact Frequency

Quiet Period

Channel Preference

Campaign Exclusions

All engagement shall respect guest consent and communication preferences maintained in the shared Guest / CRM layer.

PAGE 10 — Promotion Performance, AI Optimization & Action Center

Purpose

Provide a centralized workspace for evaluating commercial performance, identifying risks and receiving AI recommendations.

- **Promotion KPIs**

Revenue

Transactions

Units

Discount Cost

Gross Margin

Conversion

Redemption

Incremental Revenue

Average Transaction Value

Units per Transaction

Cross-Sell Rate

Upsell Rate

ROI

- **Promotion Comparison**

Promotion	Revenue	Discount	Margin	Conversion	ROI
Summer 15%	AED 186K	AED 22K	52%	18%	4.2x
Member Bundle	AED 94K	AED 8K	61%	24%	6.8x
Buy 2 Get 1	AED 72K	AED 14K	44%	31%	2.9x

- **Performance Against Control / Baseline**

Where sufficient data exists, TICVAI should distinguish between:

- **Revenue generated during a promotion**
- and

estimated incremental revenue caused by the promotion.

This is important because high promotional sales do not automatically mean the promotion created additional demand.

- **AI — PROMOTION OPTIMIZATION**

TICVAI should analyze:

Historical Promotion Performance

Product Margin

Inventory

Sales Velocity

Guest Segments

Ticket Sales

Event Attendance

Seasonality

Channel

Conversion

Basket Composition

- **Example Recommendation**

Main Gift Shop has 840 units of Summer Collection stock with declining sales velocity.

Recommendation:

- **Launch 15% Member Promotion for 7 days**

Expected:

- Revenue:
AED 72K
- Stock Reduction:
28%
- Expected Margin:
48%
- Confidence:
86%

Actions:

Simulate | Create Draft Promotion | Ignore

- **AI — PROMOTION SIMULATION**

Before activating a promotion, authorized users should be able to model its expected effect.

Example:

- **Proposed Promotion**
- 20% Off Summer Collection

Duration:

- 7 Days
 - **Current Forecast Without Promotion**
 - Units:
620
-

- Revenue:
AED 68K
- Margin:
AED 39K
- **Forecast With Promotion**
- Units:
980
- Revenue:
AED 86K
- Margin:
AED 41K

Inventory Reduction:

- +360 Units
- **Result**
- Revenue:
+26%
- Margin Value:
+5%
- Margin %:
Reduced
- Stock Reduction:
+58%

The user may then:

Modify | Save Scenario | Submit for Approval

• **AI — CROSS-SELL DISCOVERY**

TICVAI should analyze basket relationships.

Example:

Guests purchasing:

- **Logo T-Shirt**

also purchase:

- Cap — 64%
- Bag — 38%
- Water Bottle — 32%

AI Recommendation:

Add Cap as primary cross-sell recommendation across POS, App and B2C.

Estimated Monthly Revenue Impact:

- **+AED 28,000**

- **AI — PROMOTION CANNIBALIZATION**

The system should identify when a promotion appears to discount sales that likely would have happened anyway.

Example:

- **Gold Member 10% Apparel Promotion**
- Promotional Revenue:
AED 140K
- Estimated Incremental Revenue:
AED 18K
- Discount Cost:
AED 14K

AI Observation:

Most promoted purchases came from guests with historically high apparel purchase probability.

Recommendation:

Test narrower targeting rather than continuing the promotion across all Gold Members.

- **AI — INVENTORY-AWARE COMMERCE**

Board 5 shall integrate directly with Retail Board 4.

TICVAI should avoid aggressively promoting unavailable or critically low-stock products.

Example:

Campaign wants to promote:

- Limited Edition Plush
- Available:
28
- Forecast Organic Demand:
24
- Promotion Forecast:
68

Result:

- **Inventory Risk**

TICVAI recommends:

Increase stock before campaign launch.

Limit promotion to selected stores.

Reduce campaign duration.

Select alternative product.

- **ABANDONED CART & DIGITAL RECOVERY**

Where digital commerce channels support persistent carts, TICVAI may identify abandoned baskets.

Example:

Guest added:

- T-Shirt
- Cap
- Plush
- Basket:
AED 245
- No Purchase:
2 hours

If guest consent permits, TICVAI may trigger:

Your selected items are still available.

or an approved recovery offer.

- **Recovery Rules**

Configure:

Delay

Minimum Basket

Guest Segment

Product Eligibility

Offer

Communication Channel

Maximum Attempts

- **CART / CHECKOUT MERCHANDISING**

TICVAI shall support merchandising at different points of the purchase journey.

- **Product Page**
 - **You May Also Like**
 - **Cart**
 - **Complete Your Visit**
-

- **Checkout**
- **Add for AED 20**
- **Post-Purchase**
- **Collect During Your Visit**

Different placements may use different recommendation rules.

• **MEMBER & LOYALTY COMMERCE**

Board 5 shall integrate with the shared Loyalty & Membership Engine.

Examples:

- **Member Exclusive**

Gold Members:

15% off selected merchandise.

- **Points Redemption**

Use:

- 2,000 Points

to receive:

AED 20 Retail Credit.

- **Member-Only Product**

Limited Edition Collectible:

- **Gold / Platinum Members Only**

Retail shall validate eligibility but shall not maintain loyalty balances itself.

• **WALLET & ENTITLEMENT COMMERCE**

Retail shall be able to consume guest entitlements.

Example:

Guest Package includes:

- AED 100 Retail Credit

At checkout:

- Basket:
AED 180
 - Retail Entitlement:
-AED 100
 - Remaining:
AED 80
-

The entitlement shall be validated and consumed through the common TICVAI Portfolio / Wallet / Entitlement layer.

- **EVENT PRE-ORDERING**

TICVAI should support merchandise pre-order during ticket purchase or before the event.

- **Example**

Concert Ticket:

- AED 450

Add:

- Event T-Shirt:
AED 95
- VIP Poster:
AED 60

Collection:

- **Event Merchandise Counter**

The Retail order is created before arrival and inventory is reserved accordingly.

This directly connects:

- **Ticketing → Retail → Inventory → Fulfilment**
-

- **PROMOTION BUDGET & COMMERCIAL CONTROL**

Campaigns may include:

- Promotion Budget
- Maximum Discount Exposure
- Maximum Redemptions
- Revenue Target
- Margin Target

Example:

Campaign Budget:

- AED 50,000

Discount Used:

- AED 42,800

Remaining:

- AED 7,200

Status:

- **86% Consumed**

The system may warn or automatically stop a promotion according to configured policy.

- **APPROVAL ENGINE INTEGRATION**

Board 5 shall use TICVAI's common Approval Engine.

Examples:

- **Standard Promotion**
 - → Retail Manager
- **Promotion Below Margin Threshold**
 - → Retail Director + Finance
- **High Discount**
 - → Commercial Director
- **VIP / Member Benefit Change**
 - → Commercial + Loyalty Owner
- **Large Campaign Budget**
 - → Finance Approval
- **AI-Generated Promotion**
 - → Human Review + Approval

No separate Retail approval engine shall be developed.

- **PROMOTION VERSIONING & AUDIT**

Every significant commercial change shall be traceable.

Capture:

Promotion ID

Version

Created By

Changed By

Approved By

Date / Time

Previous Rule

New Rule

Products

Stores

Channels

Audience

Budget
Margin Impact
Effective Period

- **Example**
- **SUMMER-15 — Version 4**

Previous:

- 15% Off All Apparel

New:

- 15% Off Kids Apparel Only

Reason:

- Margin Optimization

Approved:

- Retail Director

Effective:

- 22 Aug 2026

• **BOARD 5 — FINAL 10 SCREENS**

#	Screen
1	Retail Commercial Command Center
2	Promotion & Offer Builder
3	Promotion Eligibility, Priority & Conflict Rules
4	Campaign & Audience Management
5	Coupons, Vouchers & Promotional Codes
6	Cross-Sell, Upsell & Recommendation Rules
7	Omnichannel Commerce & Journey Configuration
8	Ticketing, Event & Experience Commerce Integration
9	Personalized Offers & Guest Engagement
10	Promotion Performance, AI Optimization & Action Center

• **COMPLETE BOARD 5 COMMERCIAL FLOW**

- **Commercial Objective**
- ↓

Select:

- **Product / Collection**

- ↓

Define:

- **Audience**

- ↓

Define:

- **Offer / Promotion**

- ↓

Configure:

- **Eligibility + Exclusions**

- ↓

Configure:

- **Priority + Stacking**

- ↓

Select:

- **Store + Channel**

- ↓

Check:

- **Inventory Availability**

- ↓

Check:

- **Margin Impact**

- ↓

- **AI Simulation**

- Expected Sales

- Expected Margin

- Expected Inventory Impact

- Expected Conversion

- ↓

- **Approval**

- ↓

- **Schedule**

- ↓

- **Publish through Shared Promotion Engine**

- ↓

POS + Mobile POS + Kiosk + B2C + Guest App

- ↓

- Guest Interaction

- ↓
 - Transaction
 - ↓
 - Inventory + Payment + Loyalty + Finance
 - ↓
 - Performance Measurement
 - ↓
 - AI Optimization
 - ↓
 - **Improve / Extend / Pause / Stop**
-

• **ARCHITECTURAL REQUIREMENT**

The most important architectural principle for Board 5 is that TICVAI shall have **one centralized Promotion & Commerce Rules Engine**.

SoftLab should not create:

- **POS Promotions**
- separately from
- **B2C Promotions**
- separately from
- **Mobile App Promotions**
- separately from
- **Kiosk Promotions**
- separately from

Membership Discounts.

Instead:

- **Shared Promotion Engine**
- ↓
- **Eligibility**
- ↓
- **Audience**
- ↓
- **Product / SKU**
- ↓
- **Price / Benefit**
- ↓
- **Priority / Stacking**

- ↓
- **Store / Channel**
- ↓
- **Inventory Validation**
- ↓
- **Transaction Engine**
- ↓
- **Performance Measurement**

This will allow a promotion created once in TICVAI to operate consistently across every permitted sales channel.

A particularly important TICVAI differentiator should be the connection between **Ticketing + Guest + Retail + Inventory**.

For example:

TICVAI knows that 18,500 guests are attending Saturday's concert, 2,100 are VIP ticket holders, historical merchandise conversion is 21%, current event T-shirt stock is insufficient, and VIP guests historically spend 38% more on merchandise.

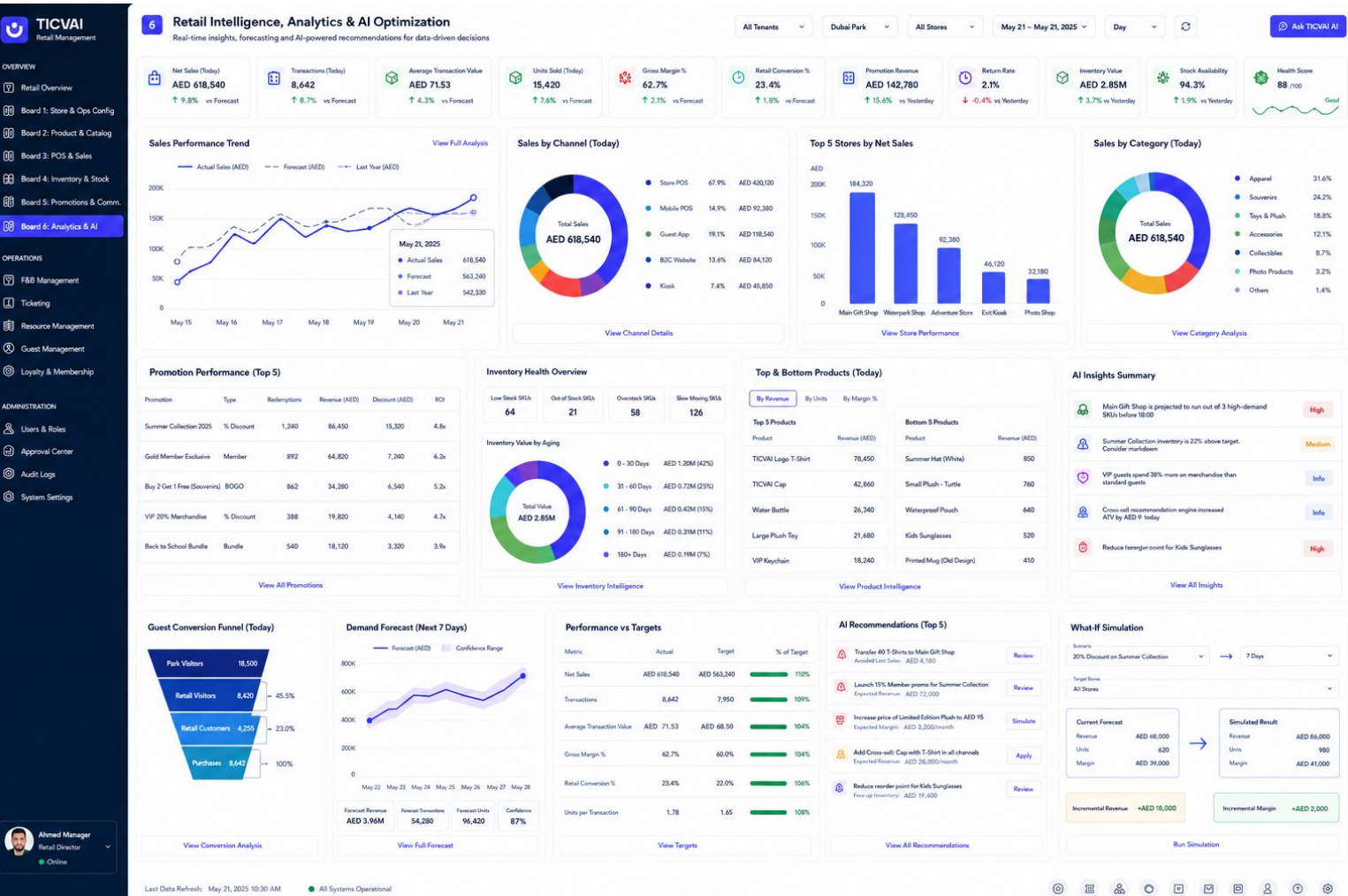
The platform can therefore recommend:

Stock → Audience → Offer → Channel → Timing → Expected Commercial Result

rather than treating Retail as a standalone shop system.

TICVAI RETAIL — BOARD 6

Retail Intelligence, Analytics, Forecasting & AI Optimization — Scope of Work



1. Purpose

The Retail Intelligence, Analytics, Forecasting & AI Optimization Board shall provide the centralized decision-intelligence layer for all TICVAI Retail operations.

Board 6 shall consolidate commercial, operational, inventory, guest, promotion, ticketing, event and financial information to provide management with a complete understanding of Retail performance and future demand.

The Board shall move beyond traditional dashboards and reporting.

The intelligence cycle shall follow:

Measure → Compare → Detect → Explain → Forecast → Recommend → Simulate → Approve → Act → Measure Outcome → Learn

The Board shall answer:

What happened?

Where did it happen?

Why did it happen?

What is likely to happen next?

What should TICVAI recommend?

What will happen if we take that action?

Board 6 shall consume information from:

- Retail POS
- Mobile POS
- Kiosk
- B2C
- Guest App
- Product & SKU Master
- Catalogs
- Pricing
- Promotions
- Retail Inventory
- Shared Inventory
- Procurement
- Guest / CRM
- Loyalty
- Membership
- Wallet & Entitlements
- Ticketing
- Attendance
- Events & Experiences
- Resource Management
- Payments
- Finance

PAGE 1 — Retail Executive Command Center

Purpose

Provide executives and Retail management with a consolidated real-time overview of commercial performance, profitability, inventory health, guest behavior and operational risks.

Global Filters

Tenant
Company
Park
Venue
Store
Store Type
Channel
Category
Brand
Collection
Date / Period

• Primary KPI Cards

Gross Sales
Net Sales
Transactions
Units Sold
Average Transaction Value
Units per Transaction
Gross Margin %
Revenue per Guest
Retail Conversion %
Member Revenue
Digital Revenue
Promotion Revenue
Inventory Value
Stock Availability %
Stock-Out Events
Return Rate
Forecast Accuracy

• Comparisons

Every applicable KPI shall support:

Today vs Yesterday

Week vs Previous Week

Month vs Previous Month

Year vs Previous Year

Actual vs Budget

Actual vs Forecast

Store vs Store

Venue vs Venue

Channel vs Channel

- **Retail Health Score**

TICVAI may calculate a configurable operational/commercial health score.

Example:

- **Main Gift Shop — 94 / 100**
- **Waterpark Shop — 87 / 100**
- **Adventure Store — 91 / 100**
- **Exit Kiosk — 76 / 100**

The score may consider:

Sales Performance

Margin

Conversion

Inventory Availability

Stock-Outs

Returns

Promotion Performance

Operational Exceptions

- **AI Executive Summary**

Example:

Retail net sales are 9.8% above forecast today. Main Gift Shop is the strongest contributor, while Exit Kiosk is 14.2% below forecast due primarily to stock-outs across five high-volume SKUs. Summer Collection inventory remains 22% above target, while Event Merchandise conversion is outperforming forecast by 18%.

Quick Actions

**Sales Analysis | Store Performance | Inventory Risks | Forecast Demand | AI
Recommendations | Ask TICVAI**

PAGE 2 — Sales, Revenue & Channel Analytics

Purpose

Provide detailed analysis of Retail sales performance across all physical and digital channels.

KPI Cards

Gross Sales
Net Sales
Transactions
Units Sold
Average Transaction Value
Units per Transaction
Discounts
Promotions
Refunds
Returns
Tax
Gross Profit
Gross Margin

- **Revenue Analysis**

Analyze by:

Tenant
Park
Venue
Store
Product
SKU
Category
Brand
Collection
Channel
Cashier
Hour
Day
Event
Guest Segment
Membership Tier
Promotion

- **Channel Analysis**

Compare:

Fixed POS
 Mobile POS
 Self-Service Kiosk
 Guest App
 B2C Website
 QR Commerce
 Event Sales
 Click & Collect

Example:

Channel	Revenue	Transactions	ATV	Units/Txn	Growth
Store POS	AED 420K	6,820	61.6	2.4	+6%
Mobile POS	AED 92K	1,180	78.0	2.8	+14%
Guest App	AED 118K	1,420	83.1	3.1	+22%
B2C	AED 84K	940	89.4	3.3	+18%
Kiosk	AED 46K	720	63.9	2.2	+9%

- **Drill-Down**

Users shall be able to navigate:

- **Company**

- → Park
- → Venue
- → Store
- → Category
- → Product
- → SKU
- → Transaction

without leaving the analytical workflow.

- **Time Analysis**

Support:

Hourly Sales
 Daily Sales
 Weekly Trends
 Monthly Trends
 Peak Period

Event Period
Opening vs Closing Period

PAGE 3 — Store & Operational Performance

Purpose

Benchmark Retail locations and identify stores requiring management attention.

• Store Scorecard

Store	Sales	AT V	Margin	Conversion	Availability	Returns	Score
Main Gift Shop	AED 184K	72	64%	28%	98%	2.1%	94
Waterpark Shop	AED 128K	61	61%	31%	94%	1.8%	87
Adventure Store	AED 92K	78	68%	22%	97%	2.4%	91
Exit Kiosk	AED 46K	39	58%	14%	86%	1.2%	76

• Store Performance Dimensions

Analyze:

- Sales
- Transactions
- ATV
- Units per Transaction
- Margin
- Retail Conversion
- Stock Availability
- Stock-Out Rate
- Return Rate
- Discount Rate
- Promotion Usage
- Employee Productivity
- Revenue per Operating Hour

• Store Ranking

Rank by:

- Revenue
- Growth
- Margin
- Conversion
- Inventory Efficiency
- Guest Spend

• Store Comparison

Authorized users should be able to select multiple stores and compare them side-by-side.

- **Root-Cause Example**

- **Exit Kiosk**

- Revenue:
-14.2% vs Forecast

Primary Factor:

- Stock Availability

Secondary Factor:

- Low ATV

Five high-volume SKUs unavailable during peak exit period.

Estimated Impact:

- **AED 8,600 Lost Sales**
-
-

PAGE 4 — Product, SKU & Merchandise Performance

Purpose

Analyze product performance using sales, profitability, demand, stock, returns and guest behavior.

- **Product KPIs**

Units Sold

Revenue

Cost

Gross Profit

Margin %

Sales Velocity

Stock Availability

Days of Supply

Stock-Out Frequency

Return Rate

Promotion Dependency

Cross-Sell Rate

Sell-Through %

Markdown %

- **Merchandise Classification**

TICVAI should classify merchandise using commercial performance.

Examples:

- **STAR**

- High Sales + High Margin

- **TRAFFIC DRIVER**

- High Sales + Lower Margin

- **OPPORTUNITY**

- Lower Sales + High Margin

- **UNDERPERFORMER**

- Low Sales + Low Margin

- **OVERSTOCK RISK**

Inventory materially exceeds expected demand.

- **Product Example**

- **TICVAI Logo T-Shirt**

- Revenue:

AED 186K

- Units:
2,140
- Margin:
62%
- Sell-Through:
74%
- Stock Availability:
97%
- Return Rate:
3.1%

Classification:

- **STAR**
- **Variant Analysis**

For apparel and similar products, performance shall be available by:

Size

Color

Gender

Style

Other Attributes

Example:

- Black / M:
Strong Demand
- Black / XL:
Low Stock
- White / S:
Slow Moving

- **AI Recommendation**

Black / XL is forecast to stock out within 1.4 days while White / S has 68 days of supply. Consider adjusting future assortment and replenishment.

PAGE 5 — Inventory, Sell-Through & Stock Intelligence

Purpose

Transform Board 4's operational stock information into management-level inventory intelligence.

KPI Cards

Inventory Value
Available Stock
Reserved Stock
In-Transit Stock
Stock Turnover
Sell-Through %
Days of Supply
Low-Stock SKUs
Stock-Out SKUs
Overstock SKUs
Slow-Moving Stock
Dead Stock
Shrinkage
Stock Adjustment Value

- **Inventory Aging**

Analyze stock by:

0–30 Days
31–60 Days
61–90 Days
91–180 Days
180+ Days

Example:

Aging	Inventory Value	%
0–30 Days	AED 1.2M	42%
31–60 Days	AED 720K	25%
61–90 Days	AED 420K	15%
91–180 Days	AED 310K	11%
180+ Days	AED 190K	7%

- **Sell-Through**

TICVAI should calculate sell-through by:

Product
SKU
Category
Brand
Collection
Store
Event
Season

- **Days of Supply**

Example:

- **Limited Edition Plush**

- Available:
280
- Average Daily Sales:
42

Days of Supply:

- **6.7 Days**
- **Slow-Moving Inventory**

Example:

- **Summer Kids Collection**

- Inventory Value:
AED 84K
- 30-Day Sales:
AED 12.4K
- Projected Days of Supply:
184

Status:

- **Overstock Risk**
- **Internal Redistribution Opportunity**

Before recommending external procurement or markdown:

Main Gift Shop requires 30 units of SKU-1842.

Waterpark Shop has projected excess of 18 units.

Adventure Store has projected excess of 22 units.

Recommendation:

Redistribute internally.

PAGE 6 — Guest, Conversion & Basket Intelligence

Purpose

Connect Retail performance with TICVAI's ticketing and guest ecosystem.

This is one of the most important differentiators of TICVAI Retail.

- **Guest KPIs**

Unique Retail Guests

Retail Conversion %

Spend per Visitor

Spend per Retail Customer

Member Spend

Non-Member Spend

VIP Spend

Repeat Retail Customers

Cross-Channel Customers

Products per Guest

- **Ticket-to-Retail Conversion**

Example:

Park Attendance:

- **18,500**

- ↓

Retail Customers:

- **4,255**

- ↓

- **Retail Conversion**

- **23%**

- **Conversion Analysis**

Analyze by:

Ticket Type

Membership

Event

Attraction

Park

Venue

Visit Time

Guest Segment

Example:

- VIP Ticket Holders:
42% Retail Conversion
- Members:
34%
- Standard Tickets:
19%

- **Basket Analysis**

Analyze:

Products per Basket

Categories per Basket

ATV

Attach Rate

Cross-Sell

Upsell

Product Associations

- **Market Basket Example**

Guests purchasing:

- **TICVAI T-Shirt**

also purchase:

- Cap — 64%
- Bag — 38%
- Water Bottle — 32%

- **Guest Journey**

Where permitted, TICVAI should analyze:

- **Ticket Purchase**

- → Visit
- → Attraction / Event
- → Retail Interaction
- → Purchase
- → Repeat Purchase

without creating a separate guest identity inside Retail.

PAGE 7 — Promotion, Pricing & Margin Intelligence

Purpose

Analyze whether pricing and promotional activity creates profitable incremental demand.

- **KPIs**

Promotion Revenue

Promotion Transactions

Discount Value

Promotion Margin

Redemption

Conversion

Incremental Revenue

Promotion ROI

Markdown Cost

Member Discount Cost

Coupon Redemption

- **Promotion Performance**

Example:

Promotion	Revenue	Discount	Margin	Conversion	ROI
Summer 15%	AED 186K	AED 22K	52%	18%	4.2x
Member Bundle	AED 94K	AED 8K	61%	24%	6.8x
Buy 2 Get 1	AED 72K	AED 14K	44%	31%	2.9x

- **Incrementality**

TICVAI should distinguish:

- **Sales during promotion**

- from

Estimated additional sales generated because of the promotion.

- **Margin Waterfall**

- **Gross Retail Sales**

- ↓

- Discounts

- ↓

- Promotions

- ↓

- Returns

- ↓

- Net Revenue
- ↓
- Product Cost
- ↓
- Inventory Adjustments / Shrinkage where applicable
- ↓
- **Contribution Margin**
- **Price Sensitivity**

Where sufficient historical data exists, TICVAI may estimate:

Demand Elasticity

Price Sensitivity

Margin Impact

Revenue Impact

- **Example**

- Current Price:
AED 95
 - Proposed:
AED 105
 - Estimated Demand:
-4.2%
 - Projected Revenue:
+6.1%
 - Projected Margin:
+9.8%
-
-

PAGE 8 — Demand Forecasting & Merchandise Planning

Purpose

Forecast future Retail demand and convert it into actionable merchandise, inventory and staffing requirements.

- **Forecast Inputs**

The engine should be capable of using:

- Advance Ticket Sales
- Attendance Forecast
- Event Schedule
- Historical Retail Sales
- Retail Conversion
- Product Sales Velocity
- Membership Visits
- Promotions
- Seasonality
- Day of Week
- Time of Day
- Store
- Product Lifecycle
- Current Stock
- Incoming Stock
- Reservations
- Weather where integrated
- Holidays
- School / Group Visits

- **Example**

- **Tomorrow — Dubai Park**

Expected Attendance:

- **16,800**

Expected Retail Conversion:

- **24%**

Expected Retail Customers:

- **4,032**

Expected Transactions:

- **4,280**

Expected Revenue:

- **AED 298,000**

- **Product Forecast**

- Logo T-Shirt:

- 1,140 units

- Cap:

- 720

- Water Bottle:

- 1,280

- Plush:

- 440

- Keychain:

- 1,860

- **Store Forecast**

- Main Gift Shop:

- AED 126K

- Waterpark Shop:

- AED 78K

- Adventure Store:

- AED 52K

- Exit Kiosk:

- AED 42K

- **Inventory Requirement**

For each SKU:

- **Forecast Demand**

- **Safety Stock**

- **Available**

- **Incoming**

- =

- **Projected Requirement**

- **Staffing Forecast**

Board 6 should integrate with TICVAI Resource Management.

Example:

- Main Gift Shop

- 18:00–21:00

- Forecast Transactions:
680
- Current Cashiers:
4

Recommended:

- **+2 Cashiers**
-
-

PAGE 9 — AI Recommendation, Simulation & Optimization Center

Purpose

Provide a centralized AI workspace for identifying commercial and operational opportunities and simulating changes before execution.

- **Recommendation Categories**

Revenue

Pricing

Promotions

Assortment

Inventory

Replenishment

Redistribution

Markdown

Cross-Sell

Upsell

Staffing

Store Operations

Guest Engagement

- **Recommendation Card**

Each recommendation should display:

- **Issue / Opportunity**
- **Business Impact**
- **Root Cause**
- **Recommended Action**
- **Confidence**
- **Expected Result**
- **Affected Stores / Products**
- **Example — Stock Redistribution**

Issue:

Main Gift Shop T-Shirt shortage.

Recommendation:

Transfer:

- 20 units from Waterpark Shop
- 20 units from Adventure Store

Expected Lost Sales Avoided:

- **AED 4,200**

Confidence:

- **92%**
- **Example — Pricing**

Increase Limited Edition Collectible from AED 150 to AED 165.

Estimated:

- Demand:
-2.8%
- Revenue:
+6.9%
- Margin:
+11.4%

- **Example — Promotion**

Create targeted 15% promotion for slow-moving Summer Collection among members visiting during the next 14 days.

Expected Stock Reduction:

- 28%

Expected Revenue:

- AED 72K

- **WHAT-IF SIMULATION**

Authorized users shall be able to simulate commercial scenarios.

Example:

What happens if we discount Summer Collection by 20% for seven days?

- **Simulation**

Current Forecast:

- 620 units
- AED 68K Revenue
- AED 39K Margin

- **Proposed Scenario**

- 980 units
- AED 86K Revenue
- AED 41K Margin

Inventory Reduction:

- +58%

Actions

Save Scenario | Modify | Compare | Submit for Approval

• MULTI-SCENARIO COMPARISON

Users should be able to compare alternative strategies.

Example:

Scenario	Revenue	Margin	Units	Ending Stock
No Action	AED 68K	AED 39K	620	840
10% Discount	AED 78K	AED 42K	790	670
20% Discount	AED 86K	AED 41K	980	480
Bundle Offer	AED 83K	AED 44K	860	610

This enables management to optimize for more than revenue alone.

PAGE 10 — AI Assistant, Alerts & Management Action Center

Purpose

Provide a conversational interface for management to query Retail data, investigate issues and initiate controlled actions.

- **Natural-Language Questions**

Authorized users should be able to ask:

How did Retail perform yesterday?

Which store is underperforming today?

Why is Exit Kiosk revenue down?

Which products are likely to run out tomorrow?

Which products are overstocked?

Which promotion generated the highest incremental margin?

Which stores have the highest return rate?

What are our top-selling products among VIP guests?

What merchandise should we prepare for Saturday's concert?

How much revenue did we lose because of stock-outs?

Which products should we transfer between stores?

Which products should we consider marking down?

- **Example Response**

Retail generated AED 618,540 today, 9.8% above forecast.

Main Gift Shop contributed most of the increase.

Exit Kiosk is 14.2% below forecast, primarily because five high-volume products were unavailable between 17:10 and 20:05.

Estimated lost revenue from stock-outs is AED 8,600.

I identified sufficient excess stock at Waterpark Shop and Adventure Store to cover 72% of the shortage without external procurement.

- **Suggested Actions**

- **View Exit Kiosk**
- **Review Stock-Outs**
- **Create Transfers**
- **View Forecast**

- **NATURAL-LANGUAGE CONTROLLED ACTIONS**

The AI Assistant should allow authorized users to propose business actions.

Example:

Transfer 30 black medium T-shirts from Waterpark Shop to Main Gift Shop tomorrow morning.

TICVAI creates:

- **Proposed Transfer**
- SKU:
TSH-BLK-M
- From:
Waterpark Shop
- To:
Main Gift Shop
- Quantity:
30
- Current Source Available:
86
- Source Forecast Demand:
32
- Destination Available:
8
- Destination Forecast Demand:
44

Expected Benefit:

Avoid approximately AED 2,850 of potential lost sales.

Actions:

Simulate | Edit | Submit for Approval | Cancel

The AI shall not bypass operational approvals.

• **AI ROOT-CAUSE ANALYSIS**

TICVAI should correlate information across modules rather than only identify symptoms.

- **Example**
- **Issue**

Exit Kiosk Revenue:

- **-14.2% vs Forecast**
- **Analysis**

Primary Cause:

- Product Availability

- ↓
- 5 Top-Selling SKUs Out of Stock
- ↓

Underlying Cause:

- Replenishment Request Submitted Late
- ↓

Underlying Cause:

- Event Demand exceeded original forecast
- **Secondary Cause**

Average Transaction Value:

- -6.1%

because cross-sell products were unavailable.

- **Estimated Impact**

Lost Revenue:

- AED 8,600

Lost Margin:

- AED 5,120

- **Recommended Actions**

Transfer stock from nearby stores.

Increase event safety stock.

Move replenishment trigger earlier.

Update event-demand forecast.

Add alternative cross-sell products.

• **AI EXPLAINABILITY**

Every material AI recommendation shall explain its reasoning.

Example:

- **Recommendation**

Transfer 40 Logo T-Shirts to Main Gift Shop.

Why?

- Available:
8
- Forecast Demand:
52

- Safety Stock:
10
- Incoming:
0
- Waterpark Excess:
22
- Adventure Excess:
28
- **Expected Result**
- Avoided Stock-Out:
44 units
- Potential Revenue Protected:
AED 4,180
- **Confidence**
- **92%**
- **Data Used**
- Current Inventory
- Reservations
- Sales Velocity
- Attendance Forecast
- Historical Conversion
- Event Schedule
- Store Demand

• FORECAST ACCURACY & MODEL PERFORMANCE

TICVAI shall measure its own forecasting accuracy.

Forecast	Predicted	Actual	Accuracy
Retail Revenue	AED 298K	AED 305K	97.7%
Transactions	4,280	4,390	97.5%
T-Shirt Units	1,140	1,188	96.0%
Retail Conversion	24%	24.8%	96.8%

Accuracy shall be available by:

- Store
- Product
- SKU
- Category

Channel

Forecast Horizon

Event

Forecast versions shall be retained so the system compares actual performance with **the forecast that existed at that point in time**, rather than recalculating historical forecasts.

• RECOMMENDATION OUTCOME TRACKING

TICVAI shall measure whether accepted recommendations achieved the expected result.

• Example

Recommendation:

Transfer 40 T-Shirts to Main Gift Shop.

Expected:

Avoid AED 4,180 Lost Sales.

• Outcome

- Units Transferred:

40

- Units Sold:

37

- Revenue Generated:

AED 3,515

- Remaining:

3

• Result

Recommendation Effectiveness:

- **84% of predicted revenue impact achieved**

This creates the intelligence loop:

Detect → Recommend → Approve → Execute → Measure → Learn

• ANOMALY DETECTION

TICVAI should proactively identify unusual patterns.

Examples:

Sudden Sales Drop

Unusual Sales Increase

Margin Deterioration

Return Spike

Excessive Discounts
 Unusual Refunds
 Stock Variance
 Shrinkage Increase
 Stock-Out Pattern
 Promotion Underperformance
 Abnormal Product Demand
 Channel Conversion Drop
 Excessive Manual Price Overrides

- **Example**

- **Return Alert — Kids T-Shirt**

Current Return Rate:

- 11.4%

Category Average:

- 3.2%

Primary Reason:

- Wrong Size

AI Observation:

Size-related returns increased after the new collection launch.

Recommendation:

Review size guide and product description before assuming a product-quality issue.

- **LOST SALES INTELLIGENCE**

TICVAI should estimate revenue lost because a product was unavailable.

- **Example**

- **Limited Edition Plush**

Unavailable:

- 14:10–18:35
- Expected Units:
68
- Actual:
4
- Estimated Lost Units:
64

Estimated Lost Revenue:

- **AED 6,080**

Estimated Lost Margin:

- **AED 3,420**

Lost-sales analytics shall support:

Store

Product

SKU

Category

Event

Time

Channel

- **MERCHANDISE / EVENT INTELLIGENCE**

TICVAI should directly correlate Retail demand with ticket and event information.

- **Example**

- Saturday Concert
- Tickets Sold:
16,800
- Forecast Attendance:
18,500
- VIP:
2,100
- Historical Merchandise Conversion:
21%

Forecast Retail Transactions:

- **3,885**
- **Product Forecast**
- Concert T-Shirt:
2,150
- Cap:
980
- Poster:
720
- VIP Collectible:
640
- **Current Inventory Risk**

Concert T-Shirt:

- Requirement:
2,150
- Available + Incoming:
1,820

Projected Shortfall:

- **330 Units**
- **Recommendation**

Allocate additional:

- **350 units**

before event opening.

• **LOCATION / TIME INTELLIGENCE**

Where permitted by the platform architecture and available data, TICVAI should identify where and when retail demand occurs.

Example:

- **Main Gift Shop**

Peak:

- 18:00–21:00

- **Waterpark Shop**

Peak:

- 14:00–17:00

- **Exit Kiosk**

Peak:

30 minutes before park closing through 45 minutes after major attraction/event release.

This information can support:

Stock Placement
POS Allocation
Staff Scheduling
Mobile POS Deployment
Promotion Timing

• **CROSS-SELL & MARKET BASKET INTELLIGENCE**

TICVAI should analyze product relationships.

Example:

- **T-Shirt Purchasers**

- Cap:
64%
- Bag:
38%
- Water Bottle:
32%
- Keychain:
18%

AI Recommendation:

Promote Logo Cap as the primary cross-sell item for T-Shirt purchases.

Expected Monthly Revenue Impact:

- **+AED 28,000**
- **Negative Relationship**

The system may also identify when products cannibalize each other.

Example:

Product A promotion increases sales but reduces Product B sales significantly.

Board 6 should present **net category impact**, not only promoted-product performance.

- **RETURN INTELLIGENCE**

Returns should be analyzed as a commercial and operational signal.

Analyze by:

Product
SKU
Variant
Store
Channel
Supplier
Return Reason
Guest Segment
Employee
Fulfilment Method

- **Example**
- **Kids Hoodie — Size M**
- Return Rate:
12.4%

- Main Reason:
Sizing
- Online Return Rate:
18%
- Store Return Rate:
5%

Possible Action:

Review digital size guidance.

This may reveal problems with:

Product Quality
Sizing
Description
Images
Fulfilment
Packaging
Supplier

• **SHRINKAGE & INVENTORY VARIANCE INTELLIGENCE**

Board 6 shall consume inventory adjustment and stock-count data from Board 4.

• **KPIs**

Shrinkage Value
Shrinkage %
Count Variance
Adjustment Value
Damage
Loss
RFID Variance

TICVAI may identify unusual patterns for investigation but shall not automatically accuse employees or guests of misconduct.

• **PROFITABILITY INTELLIGENCE**

Retail performance should not be measured only by revenue.

The system shall support:

- **Gross Sales**
 - ↓
-

- Discount
- ↓
- Promotion
- ↓
- Returns
- ↓
- Net Sales
- ↓
- Product Cost
- ↓
- Inventory Loss / Shrinkage
- ↓
- Applicable Fulfilment Cost
- ↓
- **Contribution Margin**

Analysis should be possible by:

Product
Category
Store
Channel
Campaign
Guest Segment
Event

• **ROLE-BASED INTELLIGENCE**

Board 6 shall use TICVAI's common RBAC/PBAC framework.

• **Executive / CEO**

Revenue
Margin
Growth
Forecast
Risks

AI Opportunities

• **Retail Director**

Store Performance
Product Performance

Inventory

Promotions

Forecast

- **Store Manager**

Store Sales

Stock

Staff

Returns

Operational Alerts

- **Commercial / Marketing**

Campaigns

Promotions

Conversion

Guest Segments

Cross-Sell

ROI

- **Inventory Manager**

Stock

Sell-Through

Replenishment

Transfers

Forecast

- **Finance**

Revenue

Margin

Discount

Refund

Cost

Profitability

- **CONFIGURABLE DASHBOARD BUILDER**

Authorized users shall be able to configure Retail dashboards without software development.

- **Widget Library**

Examples:

Revenue

Transactions

ATV

Units per Transaction

Margin

Conversion

Inventory

Sell-Through

Stock-Outs

Returns

Promotions

Forecast

Guest Spend

AI Insights

Alerts

Users may:

- **Add Widget**
- **Drag**
- **Resize**
- **Filter**
- **Save**
- **Share**

Dashboards may be:

Private

Role-Based

Store-Based

Venue-Based

Shared

Executive

• REPORTING, EXPORT & SCHEDULING

Analytical views shall support appropriate:

Excel Export

CSV Export

PDF Export

Scheduled Reports

Email Distribution

Saved Views

Saved Filters

- **Example**
- **Weekly Retail Executive Report**
- Every Monday — 08:00

Recipients:

- CEO
- Retail Director
- Finance Director

Contents:

Sales

Margin

Store Performance

Inventory

Promotions

Returns

Forecast

Top Risks

AI Recommendations

All distribution shall respect tenant and user permissions.

• **APPROVAL & ACTION INTEGRATION**

Board 6 shall not create a separate approval workflow.

Recommended actions shall use TICVAI's common Approval Engine.

Examples:

- **Price Change**
 - → Retail + Finance
 - **Promotion**
 - → Commercial + Finance where required
 - **Stock Transfer**
 - → Inventory
 - **Markdown**
 - → Retail / Commercial
 - **Replenishment**
 - → Inventory / Procurement
 - **Staffing Change**
 - → Resource Management
-

- **Product Discontinuation**

- → Retail Management

- **AUDITABILITY**

Material AI-driven or analytical actions shall be traceable.

Capture where applicable:

Recommendation ID

Recommendation Date

Data Context

Model / Rule Version

User

Simulation

Decision

Approval

Executed Action

Previous Value

New Value

Outcome

Timestamp

Management should be able to answer:

Why did TICVAI recommend this?

and:

Who approved it?

and:

What happened after implementation?

- **BOARD 6 — FINAL 10 SCREENS**

#	Screen
1	Retail Executive Command Center
2	Sales, Revenue & Channel Analytics
3	Store & Operational Performance
4	Product, SKU & Merchandise Performance
5	Inventory, Sell-Through & Stock Intelligence

#	Screen
6	Guest, Conversion & Basket Intelligence
7	Promotion, Pricing & Margin Intelligence
8	Demand Forecasting & Merchandise Planning
9	AI Recommendation, Simulation & Optimization Center
10	AI Assistant, Alerts & Management Action Center

- **COMPLETE BOARD 6 INTELLIGENCE FLOW**

POS + Mobile POS + Kiosk + B2C + Guest App

- ↓
- **Product + SKU + Catalog + Price**
- ↓
- **Retail Inventory + Shared Inventory**
- ↓
- **Promotion + Loyalty + Membership**
- ↓
- **Guest + Ticketing + Attendance + Events**
- ↓
- **Payment + Finance + Resource Management**
- ↓
- **TICVAI RETAIL INTELLIGENCE LAYER**
- **Collect**
- ↓
- **Normalize**
- ↓
- **Measure**
- ↓
- **Compare**
- ↓
- **Detect Anomalies**
- ↓
- **Root-Cause Analysis**
- ↓
- **Forecast**
- ↓
- **Recommend**
- ↓

- **Simulate**
 - ↓
 - **Approval**
 - ↓
 - **Execute through Source Module**
 - ↓
 - **Measure Outcome**
 - ↓
 - **Improve Future Recommendations**
-

• **ARCHITECTURAL REQUIREMENT**

Board 6 should **not be developed as ten independent reporting screens.**

It should sit on TICVAI's common **Analytics, Forecasting & AI Intelligence Layer**, the same underlying intelligence architecture we established for F&B.

The architecture should therefore be:

• **Operational Modules**

Retail + F&B + Ticketing + Inventory + Procurement + CRM + Loyalty + Resource Management + Finance

- ↓
- **Unified Data & Analytics Layer**
- ↓
- **KPI / Reporting Engine**
- ↓
- **Forecasting Engine**
- ↓
- **Anomaly Detection**
- ↓
- **Root-Cause Analysis**
- ↓
- **AI Recommendation Engine**
- ↓
- **Simulation / What-If Engine**
- ↓
- **Approval & Action Engine**
- ↓
- **Operational Modules**

- ↓

- **Outcome Measurement**

The specialized Retail Board 6 sits on top of this common intelligence architecture.

This means TICVAI does **not** need one AI engine for F&B and another AI engine for Retail. The underlying AI and analytics architecture should be shared, while each module provides its own business context, KPIs, forecasting models, rules and recommendations.

- **Final Retail Structure**

With Board 6, the Retail backend is structured as:

- **Board 1 — Store & Operational Configuration**
- **Board 2 — Product, Catalog, Pricing & Merchandising**
- **Board 3 — POS, Sales, Returns & Fulfilment**
- **Board 4 — Store Stock, Transfers & Operational Inventory**
- **Board 5 — Promotions, Omnichannel Commerce & Guest Engagement**
- **Board 6 — Intelligence, Analytics, Forecasting & AI Optimization**

This gives us a clean boundary for the next stage: **Shared Inventory & Procurement** should provide the common enterprise backbone underneath both **Retail Board 4** and the corresponding **F&B inventory/production workflows**, rather than rebuilding inventory for a third time.